Sardar Patel Institute of Technology - Department of Computer Engineering
Cryptography and Network Security Lab

Experiment 8: Evaluation of Classical and Post-Quantum Digital Signature Algorithms using Docker-Based Client-Server Communication

Lab Report

Name: Anusha Dalal UID: 2024300032 Date: 16 October 2026

Section	Submission item (Section 19)	Page content
1	Hand-drawn digital-signature diagram (Part A) with name, UID and date	Handwritten sheet
2	Table 1: execution environment + screenshots of docker ps, docker network inspect sig-net, ping test	Table 1, Screenshots 1-6
3	Screenshots of key generation, signing, transfer (incl. sender's HTTP log) and verification	Screenshots 7-12
4	Table 2 (sizes), Table 3 (tampering, original and modified text), Table 4 (timings)	Tables 2-4, Screenshots 13-19
5	Seven graphs (Section 15), Tables 5, 6 and 7	Graphs 1-7, Screenshot 20, Tables 5-7
6	Answers to all discussion questions	Handwritten pages
7	200-250 word conclusion	Handwritten page
8	Source code of common.py, keygen.py, sign.py, verify.py, tamper.py, bench.py, plot.py and the Dockerfile	Listings
9	results.csv files (sender, receiver) and UID	Tables

1. Hand-drawn digital-signature diagram (Part A)

Name: Anusha Dalal UID: 2024300032 Date: _____

Drawn with pen/pencil and ruler: Sender (Alice), insecure channel, Receiver (Bob); M, H( ), h = H(M); SK_A (dashed, never sent) and PK_A; S = Sign(SK_A, h), output (M || S); PK_A distribution; h' = H(M'), Verify(PK_A, h', S); ACCEPT / REJECT decision box; Mallory modifying M to M' -> REJECT.

[attach handwritten pages here]

2. Execution environment and Docker setup (Part C)

Table 1: Execution environment

CPU model	Cores / threads	RAM	Host OS	Docker version	cryptography version	liboqs version
12th Gen Intel(R) Core(TM) i5-1235U	12 threads (logical CPUs)	3.7 GiB	Ubuntu 24.04.5 LTS (kernel 6.6.87.2-microsoft-standard-WSL2)	29.8.2	50.0.2	0.16.0 (liboqs-python 0.16.0.1)

Host OS: Ubuntu 24.04.5 LTS. Container OS (image sig-lab:1.0, built from the Step C3 Dockerfile, FROM ubuntu:24.04): Ubuntu 24.04.5 LTS, Python 3.12.3. Container IPs on sig-net: sender 172.18.0.2, receiver 172.18.0.3.

Screenshot 1: Step C8: execution environment (CPU, RAM, OS, Docker, cryptography and liboqs versions) for Table 1

```
anudalal25@anusha:~/sig-lab$ lscpu | grep -E "Model name|^CPU\(s\)"
CPU(s):                               12
Model name:                           12th Gen Intel(R) Core(TM) i5-1235U
anudalal25@anusha:~/sig-lab$ free -h | head -2
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       838Mi        48Mi        49Mi       2.9Gi       2.9Gi
anudalal25@anusha:~/sig-lab$ lsb_release -d 2>/dev/null || grep PRETTY_NAME /etc/os-release; uname -r
Description:    Ubuntu 24.04.5 LTS
6.6.87.2-microsoft-standard-WSL2
anudalal25@anusha:~/sig-lab$ docker --version
Docker version 29.8.2, build 7fc2dff
anudalal25@anusha:~/sig-lab$ docker exec sender pip list 2>/dev/null | grep -iE "cryptography|liboqs"
cryptography    50.0.2
liboqs-python   0.16.0.1
anudalal25@anusha:~/sig-lab$ docker exec sender python3 -c "import oqs; print(oqs.oqs_version())"
liboqs-python faulthandler is disabled
0.16.0
anudalal25@anusha:~/sig-lab$ docker exec -it sender bash
root@sender:/lab# grep PRETTY_NAME /etc/os-release; python3 --version
PRETTY_NAME="Ubuntu 24.04.5 LTS"
Python 3.12.3
root@sender:/lab#
```

Screenshot 2: Step C4: the image sig-lab:1.0 has been built and is listed by docker images

```
anudalal25@anusha:~/sig-lab$ docker images sig-lab
IMAGE         ID             DISK USAGE   CONTENT SIZE   EXTRA
sig-lab:1.0   1e3b695357d4       1.23GB          302MB
anudalal25@anusha:~/sig-lab$
```

Screenshot 3: Step C5: user-defined bridge network sig-net created and listed (no containers attached yet)

```
anudalal25@anusha:~/sig-lab$ docker network create sig-net
660a45e40f472368144f72013868bd63bf10811e4c520680660132ccfbada7a1
anudalal25@anusha:~/sig-lab$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
274becbfbf0a   bridge    bridge    local
18022e9c1a7e   host      host      local
376444f39845   none      null      local
660a45e40f47   sig-net   bridge    local
anudalal25@anusha:~/sig-lab$ docker network inspect sig-net
[
    {
        "Name": "sig-net",
        "Id": "660a45e40f472368144f72013868bd63bf10811e4c520680660132ccfbada7a1",
        "Created": "2026-10-16T16:38:45.856683978Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.18.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
anudalal25@anusha:~/sig-lab$
```

Screenshot 4: Step C6: sender and receiver containers started on sig-net with STUDENT_UID; docker ps shows both Up

```
anudalal25@anusha:~/sig-lab$ export STUDENT_UID=2024300032
anudalal25@anusha:~/sig-lab$ docker run -dit --name sender   --hostname sender   --network sig-net -e STUDENT_UID=20
24300032 -v ~/sig-lab/sender:/lab   sig-lab:1.0
8dbaa8cc900f8db67c867d6a0f5745cfd3480ab997fda144e6052a7c1fb8f41b
anudalal25@anusha:~/sig-lab$ docker run -dit --name receiver --hostname receiver --network sig-net -e STUDENT_UID=20
24300032 -v ~/sig-lab/receiver:/lab sig-lab:1.0
25736136c6ad14add4570746823b8948fddad754918f90581658c99493c1bfc7
anudalal25@anusha:~/sig-lab$ docker ps
CONTAINER ID   IMAGE         COMMAND            CREATED         STATUS                  PORTS     NAMES
25736136c6ad   sig-lab:1.0   "sleep infinity"   1 second ago    Up Less than a second             receiver
8dbaa8cc900f   sig-lab:1.0   "sleep infinity"   2 seconds ago   Up 1 second                       sender
anudalal25@anusha:~/sig-lab$
```

Screenshot 5: Step C7: docker network inspect sig-net now lists both containers with their IP addresses

```
anudalal25@anusha:~/sig-lab$ docker network inspect sig-net
[
    {
        "Name": "sig-net",
        "Id": "660a45e40f472368144f72013868bd63bf10811e4c520680660132ccfbada7a1",
        "Created": "2026-10-16T16:38:45.856683978Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {
            "25736136c6ad14add4570746823b8948fddad754918f90581658c99493c1bfc7": {
                "Name": "receiver",
                "EndpointID": "3618596eb1c25c6b93aa99f10d09b1a5b66a507ee95f48182edc98a70add6b3f",
                "MacAddress": "16:21:f6:59:f6:e2",
                "IPv4Address": "172.18.0.3/16",
                "IPv6Address": ""
            },
            "8dbaa8cc900f8db67c867d6a0f5745cfd3480ab997fda144e6052a7c1fb8f41b": {
                "Name": "sender",
                "EndpointID": "14c491ab8d1141478efad1411222be6bcb362995d90d503bc17c29590ee7ccdc",
                "MacAddress": "22:8a:81:e8:d9:7e",
                "IPv4Address": "172.18.0.2/16",
                "IPv6Address": ""
            }
        },
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.18.0.0/16": {
                        "IPsInUse": 5,
                        "DynamicIPsAvailable": 65531
                    }
                }
            }
        }
    }
]
anudalal25@anusha:~/sig-lab$
```

Screenshot 6: Step C7: ping by name from sender to receiver, STUDENT_UID inside the container, liboqs mechanisms and container IPs

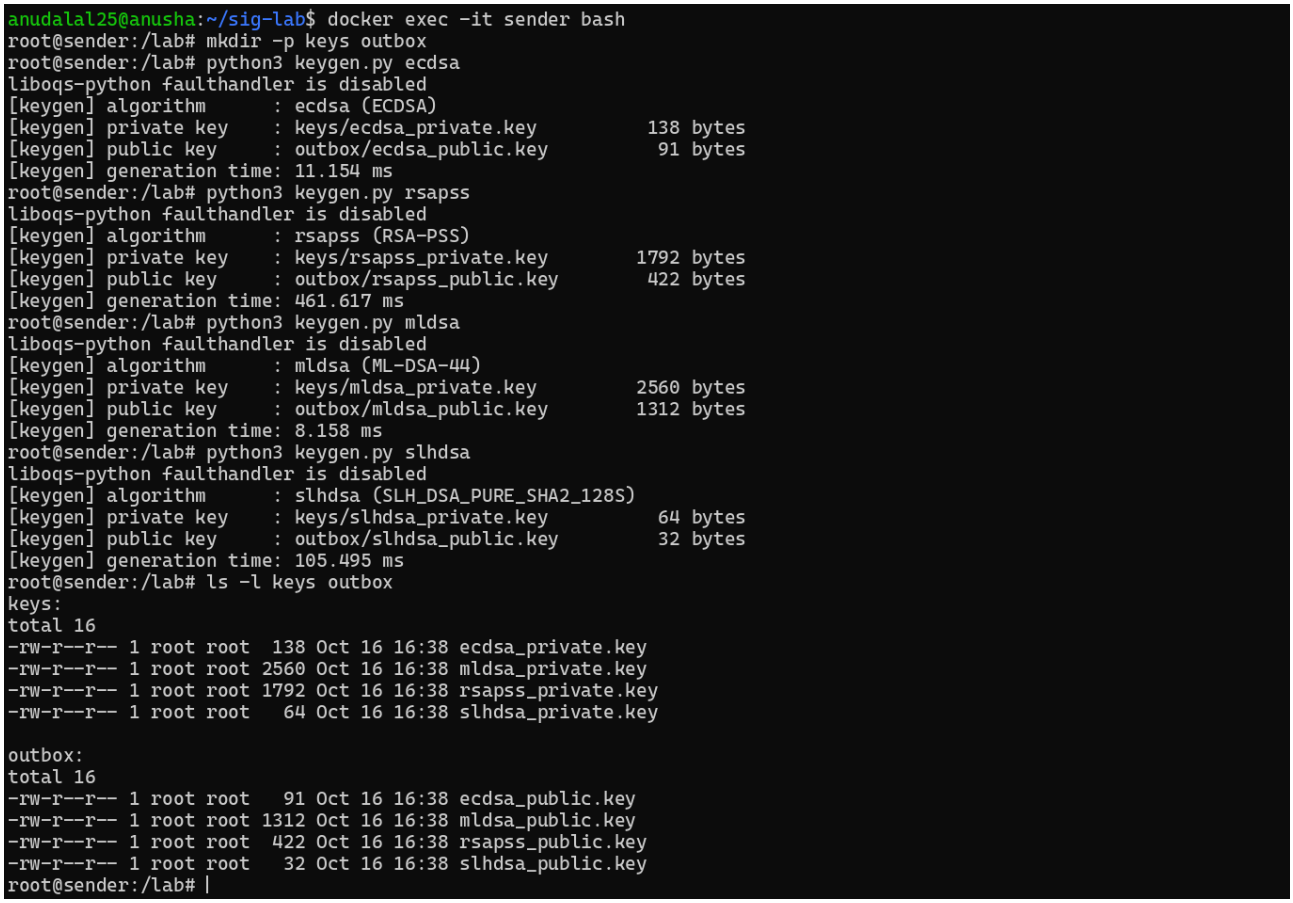


3. Key generation, signing, transfer and verification (Parts D-G)

Screenshot 7: Step D1: exact ML-DSA / SLH-DSA names in liboqs (ML-DSA-44, SLH_DSA_PURE_SHA2_128S)

```
anudalal25@anusha:~/sig-lab$ docker exec -it sender bash
root@sender:/lab# python3 -c "import oqs; print([m for m in oqs.get_enabled_sig_mechanisms() if 'ML-DSA' in m or 'SL
H' in m])"
liboqs-python faulthandler is disabled
['ML-DSA-44', 'ML-DSA-65', 'ML-DSA-87', 'SLH_DSA_PURE_SHA2_128S', 'SLH_DSA_PURE_SHA2_128F', 'SLH_DSA_PURE_SHA2_192S'
, 'SLH_DSA_PURE_SHA2_192F', 'SLH_DSA_PURE_SHA2_256S', 'SLH_DSA_PURE_SHA2_256F', 'SLH_DSA_PURE_SHAKE_128S', 'SLH_DSA_
PURE_SHAKE_128F', 'SLH_DSA_PURE_SHAKE_192S', 'SLH_DSA_PURE_SHAKE_192F', 'SLH_DSA_PURE_SHAKE_256S', 'SLH_DSA_PURE_SHA
KE_256F', 'SLH_DSA_SHA2_224_PREHASH_SHA2_128S', 'SLH_DSA_SHA2_224_PREHASH_SHA2_128F', 'SLH_DSA_SHA2_224_PREHASH_SHA2
_192S', 'SLH_DSA_SHA2_224_PREHASH_SHA2_192F', 'SLH_DSA_SHA2_224_PREHASH_SHA2_256S', 'SLH_DSA_SHA2_224_PREHASH_SHA2_2
56F', 'SLH_DSA_SHA2_224_PREHASH_SHAKE_128S', 'SLH_DSA_SHA2_224_PREHASH_SHAKE_128F', 'SLH_DSA_SHA2_224_PREHASH_SHAKE_
192S', 'SLH_DSA_SHA2_224_PREHASH_SHAKE_192F', 'SLH_DSA_SHA2_224_PREHASH_SHAKE_256S', 'SLH_DSA_SHA2_224_PREHASH_SHAKE
_256F', 'SLH_DSA_SHA2_256_PREHASH_SHA2_128S', 'SLH_DSA_SHA2_256_PREHASH_SHA2_128F', 'SLH_DSA_SHA2_256_PREHASH_SHA2_1
92S', 'SLH_DSA_SHA2_256_PREHASH_SHA2_192F', 'SLH_DSA_SHA2_256_PREHASH_SHA2_256S', 'SLH_DSA_SHA2_256_PREHASH_SHA2_256
F', 'SLH_DSA_SHA2_256_PREHASH_SHAKE_128S', 'SLH_DSA_SHA2_256_PREHASH_SHAKE_128F', 'SLH_DSA_SHA2_256_PREHASH_SHAKE_19
2S', 'SLH_DSA_SHA2_256_PREHASH_SHAKE_192F', 'SLH_DSA_SHA2_256_PREHASH_SHAKE_256S', 'SLH_DSA_SHA2_256_PREHASH_SHAKE_2
56F', 'SLH_DSA_SHA2_384_PREHASH_SHA2_128S', 'SLH_DSA_SHA2_384_PREHASH_SHA2_128F', 'SLH_DSA_SHA2_384_PREHASH_SHA2_192
S', 'SLH_DSA_SHA2_384_PREHASH_SHA2_192F', 'SLH_DSA_SHA2_384_PREHASH_SHA2_256S', 'SLH_DSA_SHA2_384_PREHASH_SHA2_256F'
, 'SLH_DSA_SHA2_384_PREHASH_SHAKE_128S', 'SLH_DSA_SHA2_384_PREHASH_SHAKE_128F', 'SLH_DSA_SHA2_384_PREHASH_SHAKE_192S
', 'SLH_DSA_SHA2_384_PREHASH_SHAKE_192F', 'SLH_DSA_SHA2_384_PREHASH_SHAKE_256S', 'SLH_DSA_SHA2_384_PREHASH_SHAKE_256
F', 'SLH_DSA_SHA2_512_PREHASH_SHA2_128S', 'SLH_DSA_SHA2_512_PREHASH_SHA2_128F', 'SLH_DSA_SHA2_512_PREHASH_SHA2_192S'
, 'SLH_DSA_SHA2_512_PREHASH_SHA2_192F', 'SLH_DSA_SHA2_512_PREHASH_SHA2_256S', 'SLH_DSA_SHA2_512_PREHASH_SHA2_256F',
'SLH_DSA_SHA2_512_PREHASH_SHAKE_128S', 'SLH_DSA_SHA2_512_PREHASH_SHAKE_128F', 'SLH_DSA_SHA2_512_PREHASH_SHAKE_192S',
 'SLH_DSA_SHA2_512_PREHASH_SHAKE_192F', 'SLH_DSA_SHA2_512_PREHASH_SHAKE_256S', 'SLH_DSA_SHA2_512_PREHASH_SHAKE_256F'
, 'SLH_DSA_SHA2_512_224_PREHASH_SHA2_128S', 'SLH_DSA_SHA2_512_224_PREHASH_SHA2_128F', 'SLH_DSA_SHA2_512_224_PREHASH_
SHA2_192S', 'SLH_DSA_SHA2_512_224_PREHASH_SHA2_192F', 'SLH_DSA_SHA2_512_224_PREHASH_SHA2_256S', 'SLH_DSA_SHA2_512_22
4_PREHASH_SHA2_256F', 'SLH_DSA_SHA2_512_224_PREHASH_SHAKE_128S', 'SLH_DSA_SHA2_512_224_PREHASH_SHAKE_128F', 'SLH_DSA
_SHA2_512_224_PREHASH_SHAKE_192S', 'SLH_DSA_SHA2_512_224_PREHASH_SHAKE_192F', 'SLH_DSA_SHA2_512_224_PREHASH_SHAKE_25
6S', 'SLH_DSA_SHA2_512_224_PREHASH_SHAKE_256F', 'SLH_DSA_SHA2_512_256_PREHASH_SHA2_128S', 'SLH_DSA_SHA2_512_256_PREH
ASH_SHA2_128F', 'SLH_DSA_SHA2_512_256_PREHASH_SHA2_192S', 'SLH_DSA_SHA2_512_256_PREHASH_SHA2_192F', 'SLH_DSA_SHA2_51
2_256_PREHASH_SHA2_256S', 'SLH_DSA_SHA2_512_256_PREHASH_SHA2_256F', 'SLH_DSA_SHA2_512_256_PREHASH_SHAKE_128S', 'SLH_
DSA_SHA2_512_256_PREHASH_SHAKE_128F', 'SLH_DSA_SHA2_512_256_PREHASH_SHAKE_192S', 'SLH_DSA_SHA2_512_256_PREHASH_SHAKE
_192F', 'SLH_DSA_SHA2_512_256_PREHASH_SHAKE_256S', 'SLH_DSA_SHA2_512_256_PREHASH_SHAKE_256F', 'SLH_DSA_SHA3_224_PREH
ASH_SHA2_128S', 'SLH_DSA_SHA3_224_PREHASH_SHA2_128F', 'SLH_DSA_SHA3_224_PREHASH_SHA2_192S', 'SLH_DSA_SHA3_224_PREHAS
H_SHA2_192F', 'SLH_DSA_SHA3_224_PREHASH_SHA2_256S', 'SLH_DSA_SHA3_224_PREHASH_SHA2_256F', 'SLH_DSA_SHA3_224_PREHASH_
SHAKE_128S', 'SLH_DSA_SHA3_224_PREHASH_SHAKE_128F', 'SLH_DSA_SHA3_224_PREHASH_SHAKE_192S', 'SLH_DSA_SHA3_224_PREHASH
_SHAKE_192F', 'SLH_DSA_SHA3_224_PREHASH_SHAKE_256S', 'SLH_DSA_SHA3_224_PREHASH_SHAKE_256F', 'SLH_DSA_SHA3_256_PREHAS
H_SHA2_128S', 'SLH_DSA_SHA3_256_PREHASH_SHA2_128F', 'SLH_DSA_SHA3_256_PREHASH_SHA2_192S', 'SLH_DSA_SHA3_256_PREHASH_
SHA2_192F', 'SLH_DSA_SHA3_256_PREHASH_SHA2_256S', 'SLH_DSA_SHA3_256_PREHASH_SHA2_256F', 'SLH_DSA_SHA3_256_PREHASH_SH
AKE_128S', 'SLH_DSA_SHA3_256_PREHASH_SHAKE_128F', 'SLH_DSA_SHA3_256_PREHASH_SHAKE_192S', 'SLH_DSA_SHA3_256_PREHASH_S
HAKE_192F', 'SLH_DSA_SHA3_256_PREHASH_SHAKE_256S', 'SLH_DSA_SHA3_256_PREHASH_SHAKE_256F', 'SLH_DSA_SHA3_384_PREHASH_
SHA2_128S', 'SLH_DSA_SHA3_384_PREHASH_SHA2_128F', 'SLH_DSA_SHA3_384_PREHASH_SHA2_192S', 'SLH_DSA_SHA3_384_PREHASH_SH
A2_192F', 'SLH_DSA_SHA3_384_PREHASH_SHA2_256S', 'SLH_DSA_SHA3_384_PREHASH_SHA2_256F', 'SLH_DSA_SHA3_384_PREHASH_SHAK
E_128S', 'SLH_DSA_SHA3_384_PREHASH_SHAKE_128F', 'SLH_DSA_SHA3_384_PREHASH_SHAKE_192S', 'SLH_DSA_SHA3_384_PREHASH_SHA
KE_192F', 'SLH_DSA_SHA3_384_PREHASH_SHAKE_256S', 'SLH_DSA_SHA3_384_PREHASH_SHAKE_256F', 'SLH_DSA_SHA3_512_PREHASH_SH
A2_128S', 'SLH_DSA_SHA3_512_PREHASH_SHA2_128F', 'SLH_DSA_SHA3_512_PREHASH_SHA2_192S', 'SLH_DSA_SHA3_512_PREHASH_SHA2
_192F', 'SLH_DSA_SHA3_512_PREHASH_SHA2_256S', 'SLH_DSA_SHA3_512_PREHASH_SHA2_256F', 'SLH_DSA_SHA3_512_PREHASH_SHAKE_
128S', 'SLH_DSA_SHA3_512_PREHASH_SHAKE_128F', 'SLH_DSA_SHA3_512_PREHASH_SHAKE_192S', 'SLH_DSA_SHA3_512_PREHASH_SHAKE
_192F', 'SLH_DSA_SHA3_512_PREHASH_SHAKE_256S', 'SLH_DSA_SHA3_512_PREHASH_SHAKE_256F', 'SLH_DSA_SHAKE_128_PREHASH_SHA
2_128S', 'SLH_DSA_SHAKE_128_PREHASH_SHA2_128F', 'SLH_DSA_SHAKE_128_PREHASH_SHA2_192S', 'SLH_DSA_SHAKE_128_PREHASH_SH
A2_192F', 'SLH_DSA_SHAKE_128_PREHASH_SHA2_256S', 'SLH_DSA_SHAKE_128_PREHASH_SHA2_256F', 'SLH_DSA_SHAKE_128_PREHASH_S
HAKE_128S', 'SLH_DSA_SHAKE_128_PREHASH_SHAKE_128F', 'SLH_DSA_SHAKE_128_PREHASH_SHAKE_192S', 'SLH_DSA_SHAKE_128_PREHA
SH_SHAKE_192F', 'SLH_DSA_SHAKE_128_PREHASH_SHAKE_256S', 'SLH_DSA_SHAKE_128_PREHASH_SHAKE_256F', 'SLH_DSA_SHAKE_256_P
REHASH_SHA2_128S', 'SLH_DSA_SHAKE_256_PREHASH_SHA2_128F', 'SLH_DSA_SHAKE_256_PREHASH_SHA2_192S', 'SLH_DSA_SHAKE_256_
PREHASH_SHA2_192F', 'SLH_DSA_SHAKE_256_PREHASH_SHA2_256S', 'SLH_DSA_SHAKE_256_PREHASH_SHA2_256F', 'SLH_DSA_SHAKE_256
_PREHASH_SHAKE_128S', 'SLH_DSA_SHAKE_256_PREHASH_SHAKE_128F', 'SLH_DSA_SHAKE_256_PREHASH_SHAKE_192S', 'SLH_DSA_SHAKE
_256_PREHASH_SHAKE_192F', 'SLH_DSA_SHAKE_256_PREHASH_SHAKE_256S', 'SLH_DSA_SHAKE_256_PREHASH_SHAKE_256F']
root@sender:/lab#
```

Screenshot 8: Step D3: keygen.py generates the ECDSA, RSA-PSS, ML-DSA and SLH-DSA key pairs and prints their sizes



Screenshot 9: Part E: sign.py creates the UID document and signs it with all four algorithms; document SHA-256 noted

```
anudalal25@anusha:~/sig-lab$ docker exec -it sender bash
root@sender:/lab# python3 sign.py ecdsa
liboqs-python faulthandler is disabled
[sign] created outbox/document.txt
[sign] algorithm     : ecdsa
[sign] document      : outbox/document.txt (59 bytes)
[sign] signature     : outbox/ecdsa.sig (71 bytes)
[sign] signing time  : 13.741 ms
root@sender:/lab# python3 sign.py rsapss
liboqs-python faulthandler is disabled
[sign] algorithm     : rsapss
[sign] document      : outbox/document.txt (59 bytes)
[sign] signature     : outbox/rsapss.sig (384 bytes)
[sign] signing time  : 5.296 ms
root@sender:/lab# python3 sign.py mldsa
liboqs-python faulthandler is disabled
[sign] algorithm     : mldsa
[sign] document      : outbox/document.txt (59 bytes)
[sign] signature     : outbox/mldsa.sig (2420 bytes)
[sign] signing time  : 1.454 ms
root@sender:/lab# python3 sign.py slhdsa
liboqs-python faulthandler is disabled
[sign] algorithm     : slhdsa
[sign] document      : outbox/document.txt (59 bytes)
[sign] signature     : outbox/slhdsa.sig (7856 bytes)
[sign] signing time  : 558.744 ms
root@sender:/lab# cat outbox/document.txt
Transfer Rs.3032 to Account 0032
Issued by UID: 2024300032
root@sender:/lab# ls -l outbox
total 40
-rw-r--r-- 1 root root   59 Oct 16 16:38 document.txt
-rw-r--r-- 1 root root   71 Oct 16 16:38 ecdsa.sig
-rw-r--r-- 1 root root   91 Oct 16 16:38 ecdsa_public.key
-rw-r--r-- 1 root root 2420 Oct 16 16:38 mldsa.sig
-rw-r--r-- 1 root root 1312 Oct 16 16:38 mldsa_public.key
-rw-r--r-- 1 root root  384 Oct 16 16:38 rsapss.sig
-rw-r--r-- 1 root root  422 Oct 16 16:38 rsapss_public.key
-rw-r--r-- 1 root root 7856 Oct 16 16:38 slhdsa.sig
-rw-r--r-- 1 root root   32 Oct 16 16:38 slhdsa_public.key
root@sender:/lab# sha256sum outbox/document.txt
d44e70dfc1a5a1f16b40f91fc9c652fc2a9e60d84dbf74435a3f4deb7cf7f388  outbox/document.txt
root@sender:/lab#
```

Screenshot 10: Part F: receiver downloads document, signatures and public keys from http://sender:8000; SHA-256 matches

```
anudalal25@anusha:~/sig-lab$ docker exec -it receiver bash
root@receiver:/lab# mkdir -p /lab/inbox && cd /lab/inbox && curl -O http://sender:8000/document.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100    59  100    59    0     0   3279      0 --:--:-- --:--:-- --:--:--  3470
root@receiver:/lab/inbox# for a in ecdsa rsapss mldsa slhdsa; do curl -sO http://sender:8000/${a}.sig; curl -sO http
://sender:8000/${a}_public.key; done
root@receiver:/lab/inbox# ls -l /lab/inbox
total 40
-rw-r--r-- 1 root root   59 Oct 16 16:39 document.txt
-rw-r--r-- 1 root root   71 Oct 16 16:39 ecdsa.sig
-rw-r--r-- 1 root root   91 Oct 16 16:39 ecdsa_public.key
-rw-r--r-- 1 root root 2420 Oct 16 16:39 mldsa.sig
-rw-r--r-- 1 root root 1312 Oct 16 16:39 mldsa_public.key
-rw-r--r-- 1 root root  384 Oct 16 16:39 rsapss.sig
-rw-r--r-- 1 root root  422 Oct 16 16:39 rsapss_public.key
-rw-r--r-- 1 root root 7856 Oct 16 16:39 slhdsa.sig
-rw-r--r-- 1 root root   32 Oct 16 16:39 slhdsa_public.key
root@receiver:/lab/inbox# sha256sum /lab/inbox/document.txt
d44e70dfc1a5a1f16b40f91fc9c652fc2a9e60d84dbf74435a3f4deb7cf7f388  /lab/inbox/document.txt
root@receiver:/lab/inbox#
```

Screenshot 11: Part F: sender's HTTP server log showing every request coming from the receiver's IP (172.18.0.3)

```
anudalal25@anusha:~/sig-lab$ docker exec -it sender bash
root@sender:/lab# cd /lab/outbox
root@sender:/lab/outbox# python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
172.18.0.3 - - [16/Oct/2026 16:39:02] "GET /document.txt HTTP/1.1" 200 -
172.18.0.3 - - [16/Oct/2026 16:39:03] "GET /ecdsa.sig HTTP/1.1" 200 -
172.18.0.3 - - [16/Oct/2026 16:39:03] "GET /ecdsa_public.key HTTP/1.1" 200 -
172.18.0.3 - - [16/Oct/2026 16:39:03] "GET /rsapss.sig HTTP/1.1" 200 -
172.18.0.3 - - [16/Oct/2026 16:39:03] "GET /rsapss_public.key HTTP/1.1" 200 -
172.18.0.3 - - [16/Oct/2026 16:39:03] "GET /mldsa.sig HTTP/1.1" 200 -
172.18.0.3 - - [16/Oct/2026 16:39:03] "GET /mldsa_public.key HTTP/1.1" 200 -
172.18.0.3 - - [16/Oct/2026 16:39:03] "GET /slhdsa.sig HTTP/1.1" 200 -
172.18.0.3 - - [16/Oct/2026 16:39:03] "GET /slhdsa_public.key HTTP/1.1" 200 -
```

Screenshot 12: Part G: verify.py on the receiver prints PASS for all four algorithms

```
anudalal25@anusha:~/sig-lab$ docker exec -it receiver bash
root@receiver:/lab# python3 verify.py ecdsa
liboqs-python faulthandler is disabled
[verify] algorithm : ecdsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/ecdsa_public.key
[verify] signature : inbox/ecdsa.sig
[verify] time      : 8.223 ms
[verify] RESULT    : PASS (VALID)
root@receiver:/lab# python3 verify.py rsapss
liboqs-python faulthandler is disabled
[verify] algorithm : rsapss
[verify] document  : inbox/document.txt
[verify] public key: inbox/rsapss_public.key
[verify] signature : inbox/rsapss.sig
[verify] time      : 1.761 ms
[verify] RESULT    : PASS (VALID)
root@receiver:/lab# python3 verify.py mldsa
liboqs-python faulthandler is disabled
[verify] algorithm : mldsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/mldsa_public.key
[verify] signature : inbox/mldsa.sig
[verify] time      : 1.161 ms
[verify] RESULT    : PASS (VALID)
root@receiver:/lab# python3 verify.py slhdsa
liboqs-python faulthandler is disabled
[verify] algorithm : slhdsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/slhdsa_public.key
[verify] signature : inbox/slhdsa.sig
[verify] time      : 0.473 ms
[verify] RESULT    : PASS (VALID)
root@receiver:/lab#
```

4. Sizes, tampering detection and timings (Parts G-I)

Table 2: Key, signature and transmission sizes (bytes)

Algorithm	Public key	Private key	Signature	Document	Transmitted (doc + sig + public key)
ECDSA	91	138	71	59	221
RSA-PSS	422	1792	384	59	865
ML-DSA	1312	2560	2420	59	3791
SLH-DSA	32	64	7856	59	7947

Sizes of the files actually written and transferred, measured with stat -c %s. Key formats as prescribed in Step D3: ECDSA and RSA-PSS private keys PKCS#8 DER, public keys SubjectPublicKeyInfo DER; ML-DSA and SLH-DSA raw keys from liboqs (these match the FIPS 204/205 sizes 1312/2560/2420 and 32/64/7856 B).

Screenshot 13: Part G: key, signature and document sizes (stat -c %s) on the sender and bytes received for Table 2

```
anudalal25@anusha:~/sig-lab$ docker exec -it sender bash
root@sender:/lab# for a in ecdsa rsapss mldsa slhdsa; do echo "$a public=$(stat -c %s outbox/${a}_public.key) privat
e=$(stat -c %s keys/${a}_private.key) sig=$(stat -c %s outbox/${a}.sig) doc=$(stat -c %s outbox/document.txt)"; done
ecdsa public=91 private=138 sig=71 doc=59
rsapss public=422 private=1792 sig=384 doc=59
mldsa public=1312 private=2560 sig=2420 doc=59
slhdsa public=32 private=64 sig=7856 doc=59
root@sender:/lab# exit
exit
anudalal25@anusha:~/sig-lab$ docker exec -it receiver bash
root@receiver:/lab# for a in ecdsa rsapss mldsa slhdsa; do echo "$a received: doc+sig+pk = $(( $(stat -c %s inbox/do
cument.txt) + $(stat -c %s inbox/${a}.sig) + $(stat -c %s inbox/${a}_public.key) )) bytes"; done
ecdsa received: doc+sig+pk = 221 bytes
rsapss received: doc+sig+pk = 865 bytes
mldsa received: doc+sig+pk = 3791 bytes
slhdsa received: doc+sig+pk = 7947 bytes
root@receiver:/lab#
```

Table 3: Verification results

Test case	ECDSA	RSA-PSS	ML-DSA	SLH-DSA	Expected
1. Original document + original signature	PASS	PASS	PASS	PASS	PASS
2. One character of document changed	FAIL	FAIL	FAIL	FAIL	FAIL
3. One bit of signature changed	FAIL	FAIL	FAIL	FAIL	FAIL
4. Correct document + signature, wrong public key	FAIL	FAIL	FAIL	FAIL	FAIL

Document	Text (tamper index = UID mod 59 = 2: 'a' (0x61) changed to '`' (0x60))
Original	Transfer Rs.3032 to Account 0032 / Issued by UID: 2024300032
Modified	Tr`nsfer Rs.3032 to Account 0032 / Issued by UID: 2024300032

Test 3 flipped bit 0 of the signature byte at index UID mod length: ECDSA byte 4 (0x65 -> 0x64), RSA-PSS byte 256 (0x19 -> 0x18), ML-DSA byte 1492 (0x78 -> 0x79), SLH-DSA byte 5232 (0xf9 -> 0xf8). Test 4: the receiver generated its own key pairs; for ECDSA an independent key from the secure random generator was used (keygen.py ecdsa --independent), because a UID-derived key would be identical to the sender's key. cmp confirmed all four receiver public keys differ from the sender's.

Screenshot 14: Part H, Tests 1-2: one character of the document changed; original passes, modified document FAILS

```
anudalal25@anusha:~/sig-lab$ docker exec -it receiver bash
root@receiver:/lab# python3 tamper.py doc inbox/document.txt
[tamper] file      : inbox/document.txt (59 bytes)
[tamper] index     : UID mod 59 = 2
[tamper] byte      : 0x61 -> 0x60 (bit 0 flipped)
[tamper] character : 'a' -> '`'
[tamper] original text:
Transfer Rs.3032 to Account 0032
Issued by UID: 2024300032
[tamper] modified text:
Tr`nsfer Rs.3032 to Account 0032
Issued by UID: 2024300032
[tamper] written   : inbox/document.txt.tampered
root@receiver:/lab# python3 verify.py ecdsa inbox/document.txt inbox/ecdsa_public.key inbox/ecdsa.sig    # Test 1: o
riginal
liboqs-python faulthandler is disabled
[verify] algorithm : ecdsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/ecdsa_public.key
[verify] signature : inbox/ecdsa.sig
[verify] time      : 2.410 ms
[verify] RESULT    : PASS (VALID)
root@receiver:/lab# python3 verify.py rsapss inbox/document.txt inbox/rsapss_public.key inbox/rsapss.sig    # Test 1
: original
liboqs-python faulthandler is disabled
[verify] algorithm : rsapss
[verify] document  : inbox/document.txt
[verify] public key: inbox/rsapss_public.key
[verify] signature : inbox/rsapss.sig
[verify] time      : 0.649 ms
[verify] RESULT    : PASS (VALID)
root@receiver:/lab# python3 verify.py mldsa inbox/document.txt inbox/mldsa_public.key inbox/mldsa.sig    # Test 1: o
riginal
liboqs-python faulthandler is disabled
[verify] algorithm : mldsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/mldsa_public.key
[verify] signature : inbox/mldsa.sig
[verify] time      : 0.427 ms
[verify] RESULT    : PASS (VALID)
root@receiver:/lab# python3 verify.py slhdsa inbox/document.txt inbox/slhdsa_public.key inbox/slhdsa.sig    # Test 1
: original
liboqs-python faulthandler is disabled
[verify] algorithm : slhdsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/slhdsa_public.key
[verify] signature : inbox/slhdsa.sig
[verify] time      : 1.091 ms
[verify] RESULT    : PASS (VALID)
root@receiver:/lab# python3 verify.py ecdsa inbox/document.txt.tampered    # Test 2: modified document
liboqs-python faulthandler is disabled
[verify] algorithm : ecdsa
[verify] document  : inbox/document.txt.tampered
[verify] public key: inbox/ecdsa_public.key
[verify] signature : inbox/ecdsa.sig
[verify] time      : 3.905 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 verify.py rsapss inbox/document.txt.tampered    # Test 2: modified document
liboqs-python faulthandler is disabled
[verify] algorithm : rsapss
[verify] document  : inbox/document.txt.tampered
[verify] public key: inbox/rsapss_public.key
[verify] signature : inbox/rsapss.sig
[verify] time      : 2.507 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 verify.py mldsa inbox/document.txt.tampered    # Test 2: modified document
liboqs-python faulthandler is disabled
[verify] algorithm : mldsa
[verify] document  : inbox/document.txt.tampered
[verify] public key: inbox/mldsa_public.key
[verify] signature : inbox/mldsa.sig
[verify] time      : 0.182 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 verify.py slhdsa inbox/document.txt.tampered    # Test 2: modified document
liboqs-python faulthandler is disabled
[verify] algorithm : slhdsa
[verify] document  : inbox/document.txt.tampered
[verify] public key: inbox/slhdsa_public.key
[verify] signature : inbox/slhdsa.sig
[verify] time      : 0.572 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab#
```

Screenshot 15: Part H, Test 3: one bit of each signature flipped at index UID mod length; all four FAIL

```
anudalal25@anusha:~/sig-lab$ docker exec -it receiver bash
root@receiver:/lab# python3 tamper.py sig inbox/ecdsa.sig
[tamper] file      : inbox/ecdsa.sig (71 bytes)
[tamper] index     : UID mod 71 = 4
[tamper] byte      : 0x65 -> 0x64 (bit 0 flipped)
[tamper] written   : inbox/ecdsa.sig.tampered
root@receiver:/lab# python3 verify.py ecdsa inbox/document.txt inbox/ecdsa_public.key inbox/ecdsa.sig.tampered    #
Test 3: modified signature
liboqs-python faulthandler is disabled
[verify] algorithm : ecdsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/ecdsa_public.key
[verify] signature : inbox/ecdsa.sig.tampered
[verify] time      : 5.337 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 tamper.py sig inbox/rsapss.sig
[tamper] file      : inbox/rsapss.sig (384 bytes)
[tamper] index     : UID mod 384 = 256
[tamper] byte      : 0x19 -> 0x18 (bit 0 flipped)
[tamper] written   : inbox/rsapss.sig.tampered
root@receiver:/lab# python3 verify.py rsapss inbox/document.txt inbox/rsapss_public.key inbox/rsapss.sig.tampered
 # Test 3: modified signature
liboqs-python faulthandler is disabled
[verify] algorithm : rsapss
[verify] document  : inbox/document.txt
[verify] public key: inbox/rsapss_public.key
[verify] signature : inbox/rsapss.sig.tampered
[verify] time      : 0.624 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 tamper.py sig inbox/mldsa.sig
[tamper] file      : inbox/mldsa.sig (2420 bytes)
[tamper] index     : UID mod 2420 = 1492
[tamper] byte      : 0x78 -> 0x79 (bit 0 flipped)
[tamper] written   : inbox/mldsa.sig.tampered
root@receiver:/lab# python3 verify.py mldsa inbox/document.txt inbox/mldsa_public.key inbox/mldsa.sig.tampered    #
Test 3: modified signature
liboqs-python faulthandler is disabled
[verify] algorithm : mldsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/mldsa_public.key
[verify] signature : inbox/mldsa.sig.tampered
[verify] time      : 0.121 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 tamper.py sig inbox/slhdsa.sig
[tamper] file      : inbox/slhdsa.sig (7856 bytes)
[tamper] index     : UID mod 7856 = 5232
[tamper] byte      : 0xf9 -> 0xf8 (bit 0 flipped)
[tamper] written   : inbox/slhdsa.sig.tampered
root@receiver:/lab# python3 verify.py slhdsa inbox/document.txt inbox/slhdsa_public.key inbox/slhdsa.sig.tampered
 # Test 3: modified signature
liboqs-python faulthandler is disabled
[verify] algorithm : slhdsa
[verify] document  : inbox/document.txt
[verify] public key: inbox/slhdsa_public.key
[verify] signature : inbox/slhdsa.sig.tampered
[verify] time      : 0.559 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab#
```

Screenshot 16: Part H, Test 4: receiver makes its own key pairs (independent ECDSA key) - verification with the wrong public key FAILS for all four

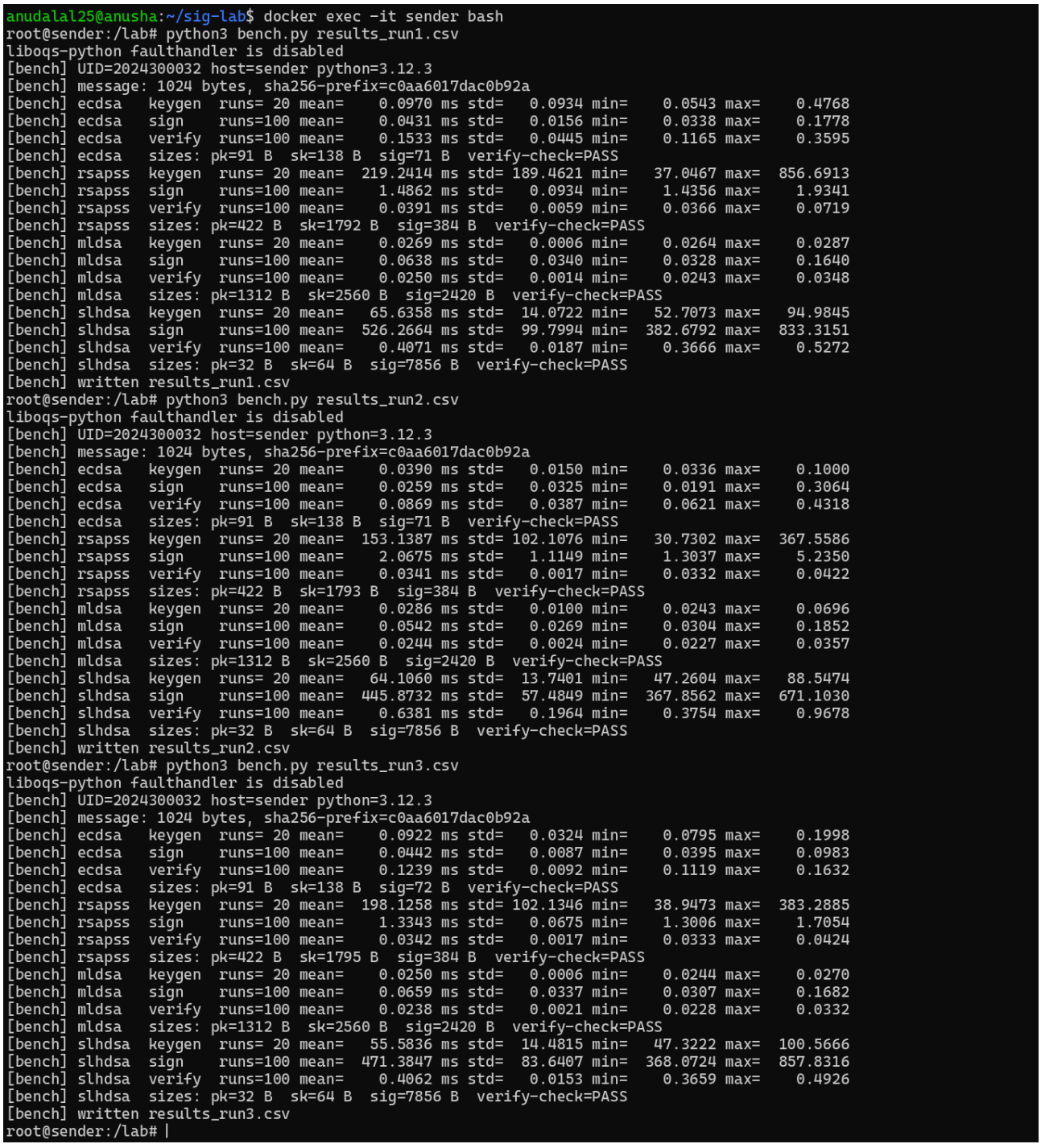
```
anudalal25@anusha:~/sig-lab$ docker exec -it receiver bash
root@receiver:/lab# mkdir -p keys outbox
root@receiver:/lab# python3 keygen.py ecdsa --independent    # Test 4: receiver makes ITS OWN key pair
liboqs-python faulthandler is disabled
[keygen] algorithm      : ecdsa (ECDSA)  [independent key from the secure RNG]
[keygen] private key    : keys/ecdsa_private.key          138 bytes
[keygen] public key     : outbox/ecdsa_public.key          91 bytes
[keygen] generation time: 0.837 ms
root@receiver:/lab# python3 verify.py ecdsa inbox/document.txt outbox/ecdsa_public.key inbox/ecdsa.sig    # Test 4:
wrong public key
liboqs-python faulthandler is disabled
[verify] algorithm : ecdsa
[verify] document  : inbox/document.txt
[verify] public key: outbox/ecdsa_public.key
[verify] signature : inbox/ecdsa.sig
[verify] time      : 2.442 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 keygen.py rsapss    # Test 4: receiver makes ITS OWN key pair
liboqs-python faulthandler is disabled
[keygen] algorithm      : rsapss (RSA-PSS)
[keygen] private key    : keys/rsapss_private.key        1793 bytes
[keygen] public key     : outbox/rsapss_public.key        422 bytes
[keygen] generation time: 160.418 ms
root@receiver:/lab# python3 verify.py rsapss inbox/document.txt outbox/rsapss_public.key inbox/rsapss.sig    # Test
4: wrong public key
liboqs-python faulthandler is disabled
[verify] algorithm : rsapss
[verify] document  : inbox/document.txt
[verify] public key: outbox/rsapss_public.key
[verify] signature : inbox/rsapss.sig
[verify] time      : 0.790 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 keygen.py mldsa    # Test 4: receiver makes ITS OWN key pair
liboqs-python faulthandler is disabled
[keygen] algorithm      : mldsa (ML-DSA-44)
[keygen] private key    : keys/mldsa_private.key         2560 bytes
[keygen] public key     : outbox/mldsa_public.key        1312 bytes
[keygen] generation time: 0.622 ms
root@receiver:/lab# python3 verify.py mldsa inbox/document.txt outbox/mldsa_public.key inbox/mldsa.sig    # Test 4:
wrong public key
liboqs-python faulthandler is disabled
[verify] algorithm : mldsa
[verify] document  : inbox/document.txt
[verify] public key: outbox/mldsa_public.key
[verify] signature : inbox/mldsa.sig
[verify] time      : 0.258 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# python3 keygen.py slhdsa    # Test 4: receiver makes ITS OWN key pair
liboqs-python faulthandler is disabled
[keygen] algorithm      : slhdsa (SLH_DSA_PURE_SHA2_128S)
[keygen] private key    : keys/slhdsa_private.key          64 bytes
[keygen] public key     : outbox/slhdsa_public.key         32 bytes
[keygen] generation time: 59.927 ms
root@receiver:/lab# python3 verify.py slhdsa inbox/document.txt outbox/slhdsa_public.key inbox/slhdsa.sig    # Test
4: wrong public key
liboqs-python faulthandler is disabled
[verify] algorithm : slhdsa
[verify] document  : inbox/document.txt
[verify] public key: outbox/slhdsa_public.key
[verify] signature : inbox/slhdsa.sig
[verify] time      : 0.530 ms
[verify] RESULT    : FAIL (INVALID)
root@receiver:/lab# for a in ecdsa rsapss mldsa slhdsa; do cmp -s inbox/${a}_public.key outbox/${a}_public.key && ec
ho "$a: receiver key IDENTICAL to sender key (not a wrong key!)" || echo "$a: receiver key differs from sender key";
 done
ecdsa: receiver key differs from sender key
rsapss: receiver key differs from sender key
mldsa: receiver key differs from sender key
slhdsa: receiver key differs from sender key
root@receiver:/lab#
```

Table 4: Timing results (milliseconds, mean ± standard deviation)

Algorithm	Key generation	Signing	Verification	Sign / Verify ratio
ECDSA	0.0922 ± 0.0324	0.0442 ± 0.0087	0.1239 ± 0.0092	0.36
RSA-PSS	198.1258 ± 102.1346	1.3343 ± 0.0675	0.0342 ± 0.0017	39.01
ML-DSA	0.0250 ± 0.0006	0.0659 ± 0.0337	0.0238 ± 0.0021	2.77
SLH-DSA	55.5836 ± 14.4815	471.3847 ± 83.6407	0.4062 ± 0.0153	1160.47

Sender container, median of three bench.py runs (results.csv = results_run3.csv); 1024-byte UID-seeded message; 5 warm-up calls, then key generation x20, signing x100, verification x100, each call timed with time.perf_counter_ns().

Screenshot 17: Part I: bench.py on the sender, run three times (results_run1/2/3.csv)



```
anudalal25@anusha:~/sig-lab$ docker exec -it sender bash
root@sender:/lab# python3 bench.py results_run1.csv
liboqs-python faulthandler is disabled
[bench] UID=2024300032 host=sender python=3.12.3
[bench] message: 1024 bytes, sha256-prefix=c0aa6017dac0b92a
[bench] ecdsa   keygen  runs= 20 mean=    0.0970 ms std=   0.0934 min=    0.0543 max=    0.4768
[bench] ecdsa   sign    runs=100 mean=    0.0431 ms std=   0.0156 min=    0.0338 max=    0.1778
[bench] ecdsa   verify  runs=100 mean=    0.1533 ms std=   0.0445 min=    0.1165 max=    0.3595
[bench] ecdsa   sizes: pk=91 B  sk=138 B  sig=71 B  verify-check=PASS
[bench] rsapss  keygen  runs= 20 mean=  219.2414 ms std= 189.4621 min=   37.0467 max=  856.6913
[bench] rsapss  sign    runs=100 mean=    1.4862 ms std=   0.0934 min=    1.4356 max=    1.9341
[bench] rsapss  verify  runs=100 mean=    0.0391 ms std=   0.0059 min=    0.0366 max=    0.0719
[bench] rsapss  sizes: pk=422 B  sk=1792 B  sig=384 B  verify-check=PASS
[bench] mldsa   keygen  runs= 20 mean=    0.0269 ms std=   0.0006 min=    0.0264 max=    0.0287
[bench] mldsa   sign    runs=100 mean=    0.0638 ms std=   0.0340 min=    0.0328 max=    0.1640
[bench] mldsa   verify  runs=100 mean=    0.0250 ms std=   0.0014 min=    0.0243 max=    0.0348
[bench] mldsa   sizes: pk=1312 B  sk=2560 B  sig=2420 B  verify-check=PASS
[bench] slhdsa  keygen  runs= 20 mean=   65.6358 ms std=  14.0722 min=   52.7073 max=   94.9845
[bench] slhdsa  sign    runs=100 mean=  526.2664 ms std=  99.7994 min=  382.6792 max=  833.3151
[bench] slhdsa  verify  runs=100 mean=    0.4071 ms std=   0.0187 min=    0.3666 max=    0.5272
[bench] slhdsa  sizes: pk=32 B  sk=64 B  sig=7856 B  verify-check=PASS
[bench] written results_run1.csv
root@sender:/lab# python3 bench.py results_run2.csv
liboqs-python faulthandler is disabled
[bench] UID=2024300032 host=sender python=3.12.3
[bench] message: 1024 bytes, sha256-prefix=c0aa6017dac0b92a
[bench] ecdsa   keygen  runs= 20 mean=    0.0390 ms std=   0.0150 min=    0.0336 max=    0.1000
[bench] ecdsa   sign    runs=100 mean=    0.0259 ms std=   0.0325 min=    0.0191 max=    0.3064
[bench] ecdsa   verify  runs=100 mean=    0.0869 ms std=   0.0387 min=    0.0621 max=    0.4318
[bench] ecdsa   sizes: pk=91 B  sk=138 B  sig=71 B  verify-check=PASS
[bench] rsapss  keygen  runs= 20 mean=  153.1387 ms std= 102.1076 min=   30.7302 max=  367.5586
[bench] rsapss  sign    runs=100 mean=    2.0675 ms std=   1.1149 min=    1.3037 max=    5.2350
[bench] rsapss  verify  runs=100 mean=    0.0341 ms std=   0.0017 min=    0.0332 max=    0.0422
[bench] rsapss  sizes: pk=422 B  sk=1793 B  sig=384 B  verify-check=PASS
[bench] mldsa   keygen  runs= 20 mean=    0.0286 ms std=   0.0100 min=    0.0243 max=    0.0696
[bench] mldsa   sign    runs=100 mean=    0.0542 ms std=   0.0269 min=    0.0304 max=    0.1852
[bench] mldsa   verify  runs=100 mean=    0.0244 ms std=   0.0024 min=    0.0227 max=    0.0357
[bench] mldsa   sizes: pk=1312 B  sk=2560 B  sig=2420 B  verify-check=PASS
[bench] slhdsa  keygen  runs= 20 mean=   64.1060 ms std=  13.7401 min=   47.2604 max=   88.5474
[bench] slhdsa  sign    runs=100 mean=  445.8732 ms std=  57.4849 min=  367.8562 max=  671.1030
[bench] slhdsa  verify  runs=100 mean=    0.6381 ms std=   0.1964 min=    0.3754 max=    0.9678
[bench] slhdsa  sizes: pk=32 B  sk=64 B  sig=7856 B  verify-check=PASS
[bench] written results_run2.csv
root@sender:/lab# python3 bench.py results_run3.csv
liboqs-python faulthandler is disabled
[bench] UID=2024300032 host=sender python=3.12.3
[bench] message: 1024 bytes, sha256-prefix=c0aa6017dac0b92a
[bench] ecdsa   keygen  runs= 20 mean=    0.0922 ms std=   0.0324 min=    0.0795 max=    0.1998
[bench] ecdsa   sign    runs=100 mean=    0.0442 ms std=   0.0087 min=    0.0395 max=    0.0983
[bench] ecdsa   verify  runs=100 mean=    0.1239 ms std=   0.0092 min=    0.1119 max=    0.1632
[bench] ecdsa   sizes: pk=91 B  sk=138 B  sig=72 B  verify-check=PASS
[bench] rsapss  keygen  runs= 20 mean=  198.1258 ms std= 102.1346 min=   38.9473 max=  383.2885
[bench] rsapss  sign    runs=100 mean=    1.3343 ms std=   0.0675 min=    1.3006 max=    1.7054
[bench] rsapss  verify  runs=100 mean=    0.0342 ms std=   0.0017 min=    0.0333 max=    0.0424
[bench] rsapss  sizes: pk=422 B  sk=1795 B  sig=384 B  verify-check=PASS
[bench] mldsa   keygen  runs= 20 mean=    0.0250 ms std=   0.0006 min=    0.0244 max=    0.0270
[bench] mldsa   sign    runs=100 mean=    0.0659 ms std=   0.0337 min=    0.0307 max=    0.1682
[bench] mldsa   verify  runs=100 mean=    0.0238 ms std=   0.0021 min=    0.0228 max=    0.0332
[bench] mldsa   sizes: pk=1312 B  sk=2560 B  sig=2420 B  verify-check=PASS
[bench] slhdsa  keygen  runs= 20 mean=   55.5836 ms std=  14.4815 min=   47.3222 max=  100.5666
[bench] slhdsa  sign    runs=100 mean=  471.3847 ms std=  83.6407 min=  368.0724 max=  857.8316
[bench] slhdsa  verify  runs=100 mean=    0.4062 ms std=   0.0153 min=    0.3659 max=    0.4926
[bench] slhdsa  sizes: pk=32 B  sk=64 B  sig=7856 B  verify-check=PASS
[bench] written results_run3.csv
root@sender:/lab#
```

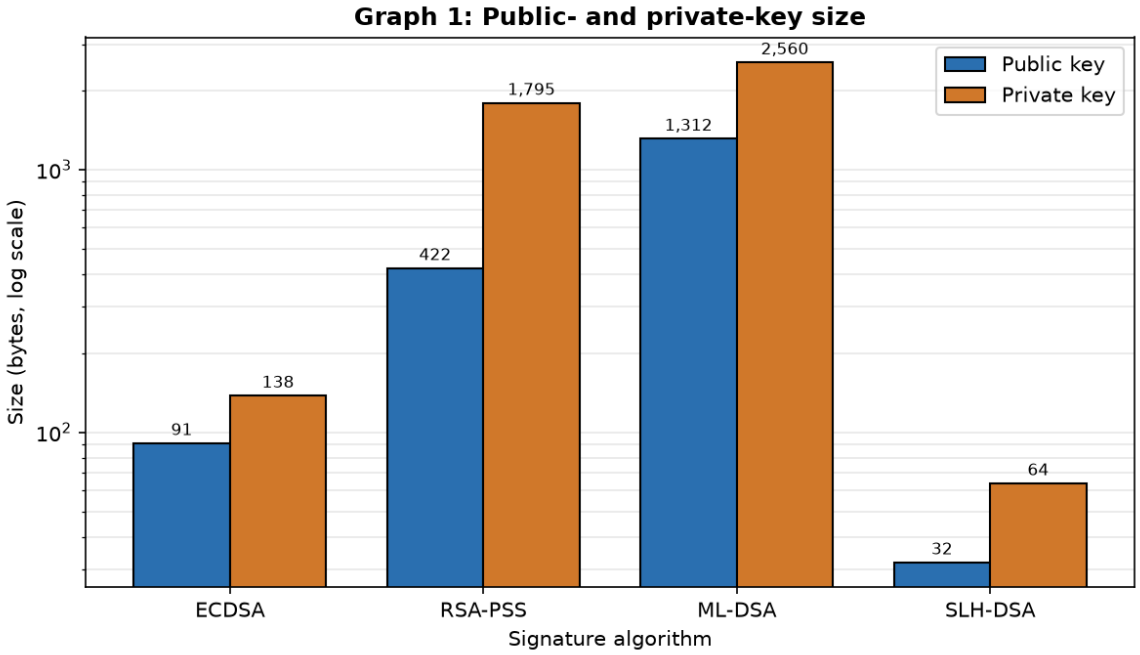
Screenshot 18: Part I: the median run (run 3) is kept as results.csv on the sender

```
anudalal25@anusha:~/sig-lab$ docker exec -it sender bash
root@sender:/lab# for r in 1 2 3; do echo "run $r: sum of mean_ms = $(awk -F, 'NR>1{s+=$5} END{printf "%.2f", s}' re
sults_run$r.csv)"; done
run 1: sum of mean_ms = 813.49
run 2: sum of mean_ms = 666.12
run 3: sum of mean_ms = 727.24
root@sender:/lab# cp results_run3.csv results.csv && cat results.csv
uid,algorithm,operation,runs,mean_ms,std_ms,min_ms,max_ms,public_key_bytes,private_key_bytes,signature_bytes
2024300032,ecdsa,keygen,20,0.0922,0.0324,0.0795,0.1998,91,138,72
2024300032,ecdsa,sign,100,0.0442,0.0087,0.0395,0.0983,91,138,72
2024300032,ecdsa,verify,100,0.1239,0.0092,0.1119,0.1632,91,138,72
2024300032,rsapss,keygen,20,198.1258,102.1346,38.9473,383.2885,422,1795,384
2024300032,rsapss,sign,100,1.3343,0.0675,1.3006,1.7054,422,1795,384
2024300032,rsapss,verify,100,0.0342,0.0017,0.0333,0.0424,422,1795,384
2024300032,mldsa,keygen,20,0.0250,0.0006,0.0244,0.0270,1312,2560,2420
2024300032,mldsa,sign,100,0.0659,0.0337,0.0307,0.1682,1312,2560,2420
2024300032,mldsa,verify,100,0.0238,0.0021,0.0228,0.0332,1312,2560,2420
2024300032,slhdsa,keygen,20,55.5836,14.4815,47.3222,100.5666,32,64,7856
2024300032,slhdsa,sign,100,471.3847,83.6407,368.0724,857.8316,32,64,7856
2024300032,slhdsa,verify,100,0.4062,0.0153,0.3659,0.4926,32,64,7856
root@sender:/lab#
```

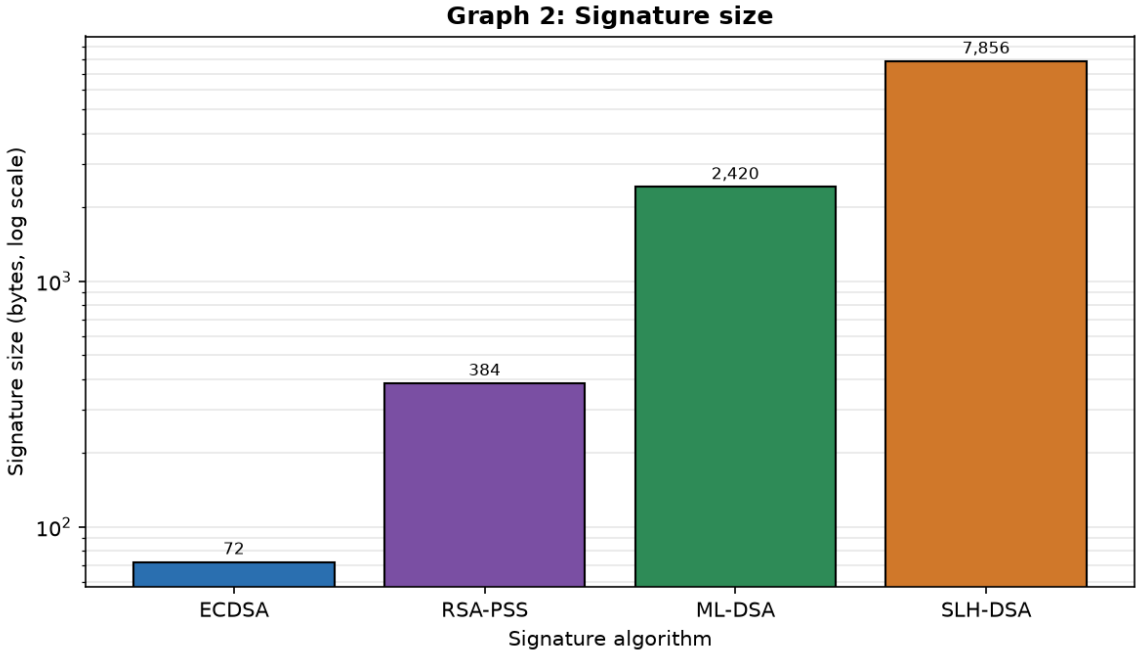
Screenshot 19: Part I: bench.py repeated on the receiver container (receiver results.csv)

```
anudalal25@anusha:~/sig-lab$ docker exec -it receiver bash
root@receiver:/lab# python3 bench.py results.csv
liboqs-python faulthandler is disabled
[bench] UID=2024300032 host=receiver python=3.12.3
[bench] message: 1024 bytes, sha256-prefix=c0aa6017dac0b92a
[bench] ecdsa   keygen  runs= 20 mean=    0.0324 ms std=   0.0054 min=    0.0302 max=    0.0544
[bench] ecdsa   sign    runs=100 mean=    0.0194 ms std=   0.0015 min=    0.0188 max=    0.0326
[bench] ecdsa   verify  runs=100 mean=    0.0597 ms std=   0.0124 min=    0.0532 max=    0.1340
[bench] ecdsa   sizes: pk=91 B  sk=138 B  sig=70 B  verify-check=PASS
[bench] rsapss  keygen  runs= 20 mean=  115.7305 ms std=  76.3442 min=   49.8620 max=  323.9396
[bench] rsapss  sign    runs=100 mean=    1.3643 ms std=   0.1119 min=    1.3034 max=    2.0226
[bench] rsapss  verify  runs=100 mean=    0.0347 ms std=   0.0026 min=    0.0332 max=    0.0489
[bench] rsapss  sizes: pk=422 B  sk=1794 B  sig=384 B  verify-check=PASS
[bench] mldsa   keygen  runs= 20 mean=    0.0279 ms std=   0.0053 min=    0.0248 max=    0.0490
[bench] mldsa   sign    runs=100 mean=    0.0632 ms std=   0.0363 min=    0.0308 max=    0.2783
[bench] mldsa   verify  runs=100 mean=    0.0253 ms std=   0.0042 min=    0.0230 max=    0.0537
[bench] mldsa   sizes: pk=1312 B  sk=2560 B  sig=2420 B  verify-check=PASS
[bench] slhdsa  keygen  runs= 20 mean=   64.1955 ms std=  23.6940 min=   47.6752 max=  132.3601
[bench] slhdsa  sign    runs=100 mean=  488.9761 ms std=  68.7288 min=  376.9210 max=  809.7748
[bench] slhdsa  verify  runs=100 mean=    0.3776 ms std=   0.0156 min=    0.3334 max=    0.4416
[bench] slhdsa  sizes: pk=32 B  sk=64 B  sig=7856 B  verify-check=PASS
[bench] written results.csv
root@receiver:/lab#
```

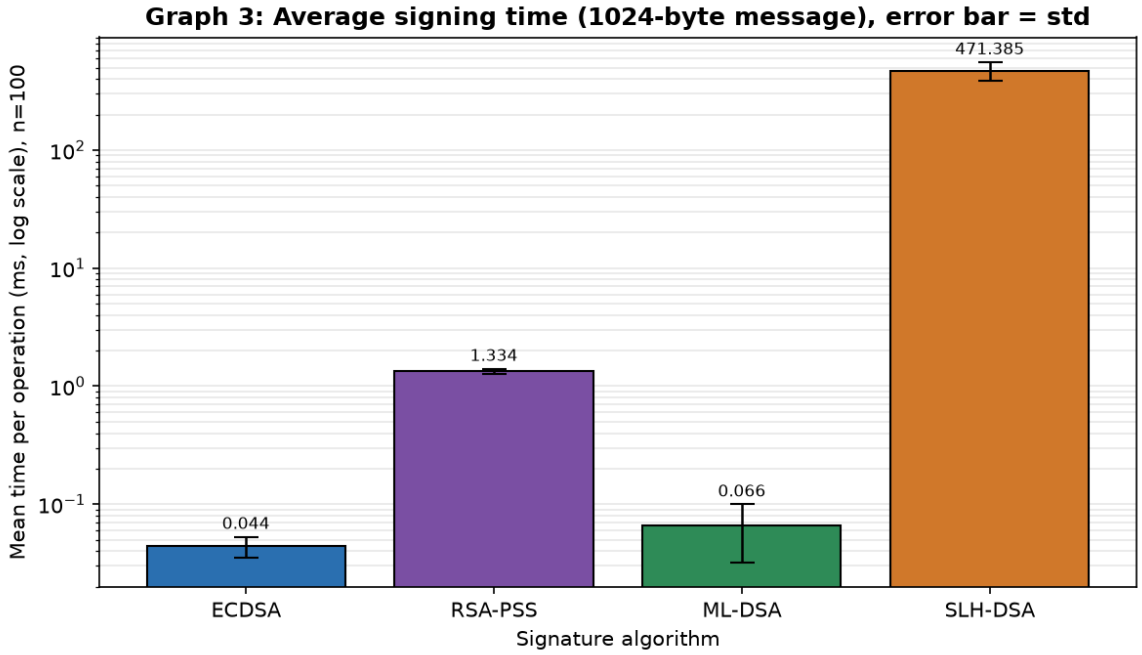
5. Graphical analysis (Section 15) and comparative analysis (Part J)



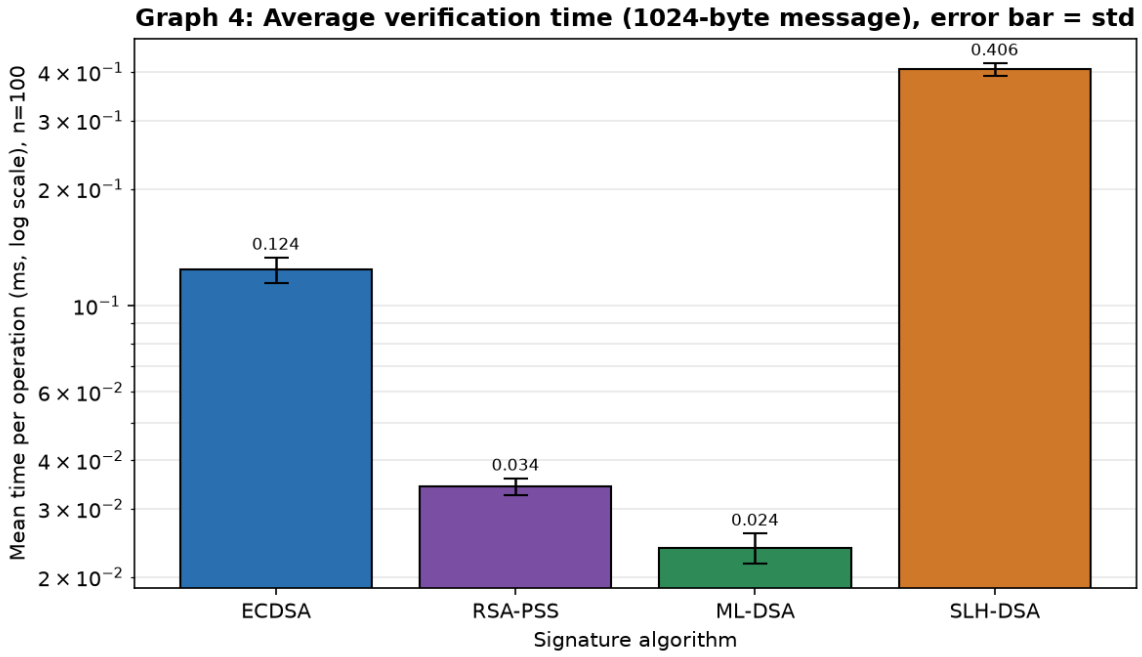
Graph 1: Public-key and private-key size (bytes, log scale). Environment: CPU: 12th Gen Intel(R) Core(TM) i5-1235U (12 threads), RAM 3.7Gi; Host: Ubuntu 24.04.5 LTS, kernel 6.6.87.2-microsoft-standard-WSL2, Docker version 29.8.2; Container: Ubuntu 24.04.5 LTS, cryptography 50.0.2, liboqs 0.16.0. UID 2024300032.



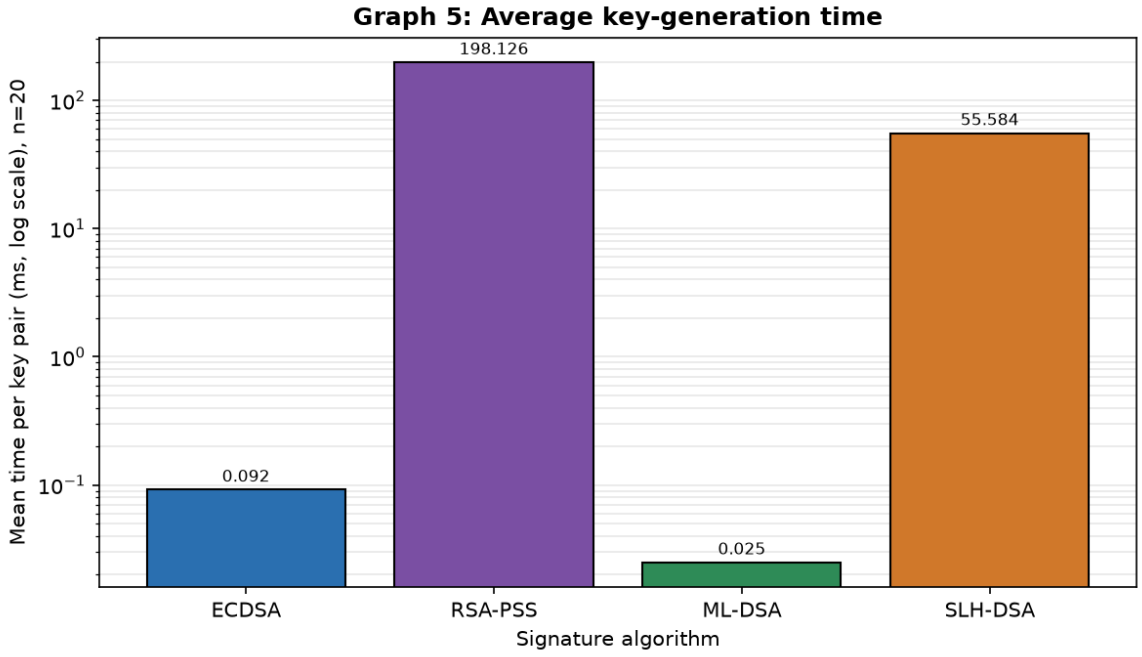
Graph 2: Signature size (bytes, log scale). Environment: CPU: 12th Gen Intel(R) Core(TM) i5-1235U (12 threads), RAM 3.7Gi; Host: Ubuntu 24.04.5 LTS, kernel 6.6.87.2-microsoft-standard-WSL2, Docker version 29.8.2; Container: Ubuntu 24.04.5 LTS, cryptography 50.0.2, liboqs 0.16.0. UID 2024300032.



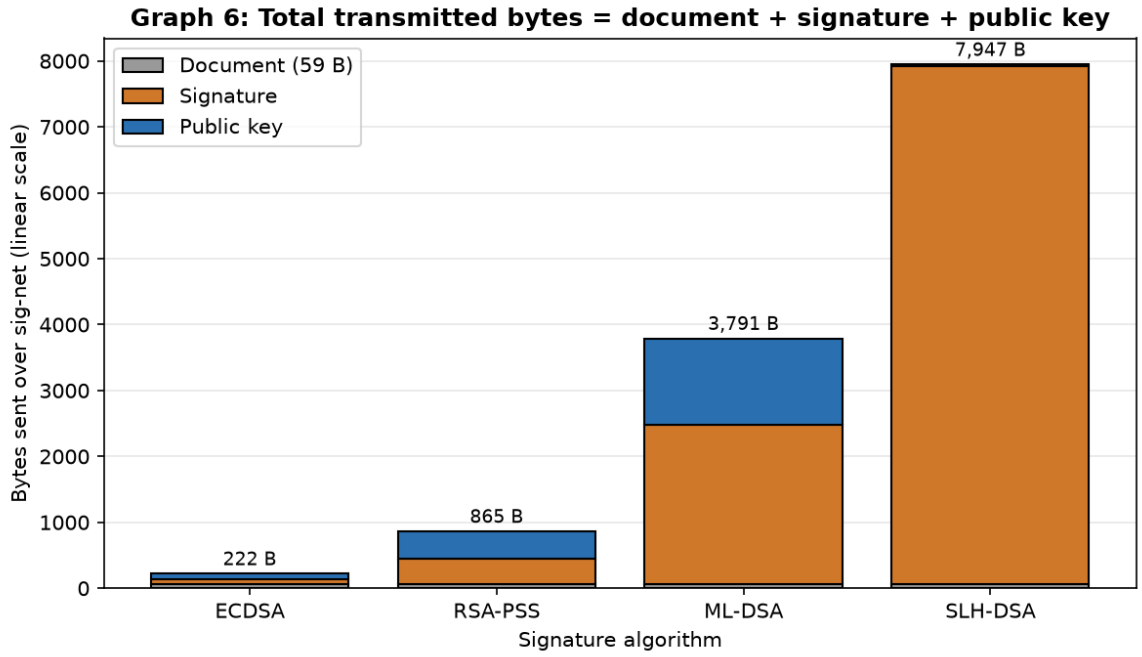
Graph 3: Average signing time (ms, log scale); error bar = standard deviation, n = 100. Environment: CPU: 12th Gen Intel(R) Core(TM) i5-1235U (12 threads), RAM 3.7Gi; Host: Ubuntu 24.04.5 LTS, kernel 6.6.87.2-microsoft-standard-WSL2, Docker version 29.8.2; Container: Ubuntu 24.04.5 LTS, cryptography 50.0.2, liboqs 0.16.0. UID 2024300032.



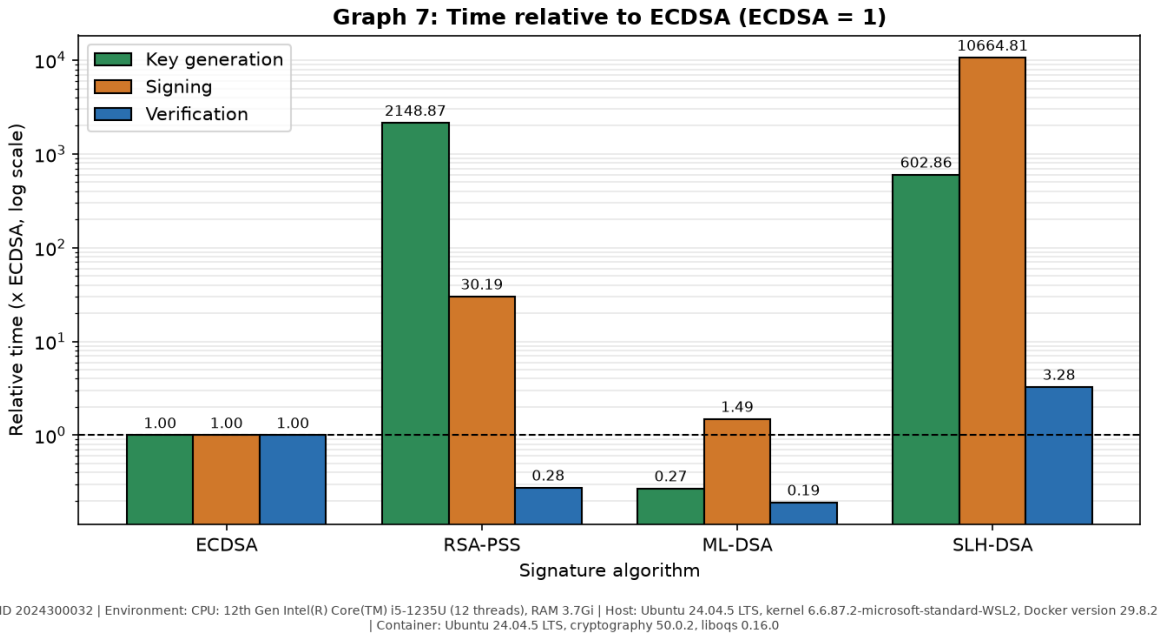
Graph 4: Average verification time (ms, log scale); error bar = standard deviation, n = 100. Environment: CPU: 12th Gen Intel(R) Core(TM) i5-1235U (12 threads), RAM 3.7Gi; Host: Ubuntu 24.04.5 LTS, kernel 6.6.87.2-microsoft-standard-WSL2, Docker version 29.8.2; Container: Ubuntu 24.04.5 LTS, cryptography 50.0.2, liboqs 0.16.0. UID 2024300032.



Graph 5: Average key-generation time (ms, log scale), n = 20. Environment: CPU: 12th Gen Intel(R) Core(TM) i5-1235U (12 threads), RAM 3.7Gi; Host: Ubuntu 24.04.5 LTS, kernel 6.6.87.2-microsoft-standard-WSL2, Docker version 29.8.2; Container: Ubuntu 24.04.5 LTS, cryptography 50.0.2, liboqs 0.16.0. UID 2024300032.



Graph 6: Total transmitted bytes = document + signature + public key. Environment: CPU: 12th Gen Intel(R) Core(TM) i5-1235U (12 threads), RAM 3.7Gi; Host: Ubuntu 24.04.5 LTS, kernel 6.6.87.2-microsoft-standard-WSL2, Docker version 29.8.2; Container: Ubuntu 24.04.5 LTS, cryptography 50.0.2, liboqs 0.16.0. UID 2024300032.



Graph 7 (extra): Sign, verify and key-generation time relative to ECDSA (ECDSA = 1, log scale). Environment: CPU: 12th Gen Intel(R) Core(TM) i5-1235U (12 threads), RAM 3.7Gi; Host: Ubuntu 24.04.5 LTS, kernel 6.6.87.2-microsoft-standard-WSL2, Docker version 29.8.2; Container: Ubuntu 24.04.5 LTS, cryptography 50.0.2, liboqs 0.16.0. UID 2024300032.

Screenshot 20: Section 15: plot.py draws the seven graphs; docker cp copies graphs/ to the host

```
anudalal25@anusha:~/sig-lab$ docker exec -it sender bash
root@sender:/lab# python3 plot.py results.csv outbox/document.txt
[plot] wrote graphs/graph1_key_sizes.png
[plot] wrote graphs/graph2_signature_size.png
[plot] wrote graphs/graph3_signing_time.png
[plot] wrote graphs/graph4_verification_time.png
[plot] wrote graphs/graph5_keygen_time.png
[plot] wrote graphs/graph6_transmitted_bytes.png
[plot] wrote graphs/graph7_relative_to_ecdsa.png
root@sender:/lab# exit
exit
anudalal25@anusha:~/sig-lab$ docker cp sender:/lab/graphs ./graphs
anudalal25@anusha:~/sig-lab$ ls -l graphs
total 536
-rw-r--r-- 1 anudalal25 anudalal25 70651 Oct 16 16:43 graph1_key_sizes.png
-rw-r--r-- 1 anudalal25 anudalal25 63036 Oct 16 16:43 graph2_signature_size.png
-rw-r--r-- 1 anudalal25 anudalal25 74949 Oct 16 16:43 graph3_signing_time.png
-rw-r--r-- 1 anudalal25 anudalal25 80915 Oct 16 16:43 graph4_verification_time.png
-rw-r--r-- 1 anudalal25 anudalal25 72286 Oct 16 16:43 graph5_keygen_time.png
-rw-r--r-- 1 anudalal25 anudalal25 82052 Oct 16 16:43 graph6_transmitted_bytes.png
-rw-r--r-- 1 anudalal25 anudalal25 89924 Oct 16 16:43 graph7_relative_to_ecdsa.png
anudalal25@anusha:~/sig-lab$
```

Table 5: Relative comparison with ECDSA as the baseline (measured value / ECDSA value)

Parameter	ECDSA	RSA-PSS	ML-DSA	SLH-DSA
Public key size	1.0	4.64	14.42	0.35
Signature size	1.0	5.41	34.08	110.65
Signing time	1.0	30.19	1.49	10664.81
Verification time	1.0	0.28	0.19	3.28
Key generation time	1.0	2148.87	0.27	602.86
Total transmitted bytes	1.0	3.91	17.15	35.96

Table 6: Property comparison

Property	ECDSA	RSA-PSS	ML-DSA	SLH-DSA
Standard	FIPS 186-5	RFC 8017, FIPS 186-5	FIPS 204 (2024)	FIPS 205 (2024)
Security parameter / category	P-256 (secp256r1), ~128-bit	RSA-3072, ~128-bit	ML-DSA-44, NIST category 2	SLH-DSA-SHA2-128s, NIST category 1
Mathematical basis	Elliptic-curve discrete logarithm	Integer factorisation	Module lattices (Module-LWE / SIS)	Hash functions only (hypertree of Merkle / FORS trees)
Resistant to Shor's algorithm?	No	No	Yes	Yes
Public key size (small / medium / large)	Small (91 B)	Medium (422 B)	Large (1312 B)	Small (32 B)
Signature size	Small (71 B)	Medium (384 B)	Large (2420 B)	Very large (7856 B)
Fastest operation	Signing (0.044 ms)	Verification (0.034 ms)	Verification (0.024 ms)	Verification (0.406 ms)
Slowest operation	Verification (0.124 ms)	Key generation (198.1 ms)	Signing (0.066 ms)	Signing (471.4 ms)
Maturity / deployment	Very mature; TLS, SSH, code signing, smart cards	Very mature (decades); PKI, TLS, document signing	New (2024); being rolled out, often in hybrid mode	New (2024), conservative SPHINCS+ design; firmware, root keys
Main advantage	Smallest keys and signatures, fast signing	Very fast verification, universal support	Quantum-safe and fast (fastest verification measured)	Quantum-safe, relies only on hash functions; 32 B public key
Main disadvantage	Broken by Shor; needs secure nonces	Broken by Shor; slow key generation, larger keys	Large public key and signature	Very slow signing, very large signature

Table 7: Which algorithm would you choose?

Scenario	Chosen algorithm	Justification
Signing a firmware image for a device that stays in the field for 15 years	SLH-DSA	Must outlive quantum computers; hash-only security is the most conservative, signing is rare and verification was cheap in our tests.
Authenticating a short sensor message on a very low-bandwidth link	ECDSA	Smallest overhead (71 B signature, 91 B key) and the data is short-lived.
Signing a TLS server certificate in 2026 on a general web server	ECDSA (P-256)	Universally supported, tiny handshake; certificates live < 1 year, with a planned move to hybrid ECDSA + ML-DSA.
Root key of a certificate authority that must remain trusted for 30 years	SLH-DSA	Rarely signs, so slow signing and large signatures are acceptable; security rests only on hash functions.
High-volume server that verifies millions of signatures per second	ML-DSA	Fastest verification in our measurements (0.024 ms per signature on this setup) and quantum-safe.

Timings in Tables 4-7 are those measured in this experiment's containers on the machine in Table 1; absolute values depend on the hardware and library build, the relative ordering is what the choices rely on.

6. Answers to the discussion questions

Questions 1-11 (Section 17), handwritten.

[attach handwritten pages here]

7. Final conclusion (200-250 words)

Handwritten conclusion (Section 18).

[attach handwritten pages here]

8. Source code

Dockerfile

```
FROM ubuntu:24.04
ENV DEBIAN_FRONTEND=noninteractive
RUN apt-get update && apt-get install -y --no-install-recommends \
    python3 python3-pip python3-venv git cmake ninja-build build-essential \
    libssl-dev curl wget nano iproute2 iputils-ping tcpdump ca-certificates \
    && rm -rf /var/lib/apt/lists/*
RUN python3 -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip install --no-cache-dir cryptography matplotlib pandas liboqs-python
# The first import makes liboqs-python download and build liboqs (a few minutes)
RUN python3 -c "import oqs; print(oqs.oqs_version())"
WORKDIR /lab
CMD ["sleep", "infinity"]
```

common.py

```
"""common.py - shared helpers for the digital-signature lab (sender and receiver).

Contents (Step D2):
  * STUDENT_UID read from the environment (must be exactly 10 digits)
  * ALGS: lab name -> library name / parameters
  * document text, ECDSA private scalar, benchmark message, tamper index (Section 5)
  * file-path helpers and a millisecond timing helper
"""
import hashlib
import os
import random
import sys
import time

# ---------------------------------------------------------------- UID
def get_uid():
    uid = os.environ.get("STUDENT_UID", "").strip()
    if not uid:
        sys.exit("ERROR: environment variable STUDENT_UID is not set "
                 "(start the container with -e STUDENT_UID=<your 10-digit UID>)")
    if not (uid.isdigit() and len(uid) == 10):
        sys.exit(f"ERROR: STUDENT_UID must be exactly 10 digits, got {uid!r}")
    return uid


UID = get_uid()

# ---------------------------------------------------------------- algorithms
# Strings for the PQ schemes are the exact names printed by Step D1 on this system.
ALGS = {
    "ecdsa":  {"family": "classical", "label": "ECDSA",   "curve": "SECP256R1", "hash": "SHA256"},
    "rsapss": {"family": "classical", "label": "RSA-PSS", "key_size": 3072, "e": 65537,
               "hash": "SHA256", "salt_length": 32},
    "mldsa":  {"family": "pq", "label": "ML-DSA",  "oqs_name": "ML-DSA-44"},
    "slhdsa": {"family": "pq", "label": "SLH-DSA", "oqs_name": "SLH_DSA_PURE_SHA2_128S"},
}
ALG_ORDER = ["ecdsa", "rsapss", "mldsa", "slhdsa"]


def check_alg(alg):
    if alg not in ALGS:
        sys.exit(f"ERROR: unknown algorithm {alg!r}; choose one of: {', '.join(ALG_ORDER)}")
    return alg


# ---------------------------------------------------------------- UID-derived values (Section 5)
# Order n of the P-256 group
P256_N = int("FFFFFFFF00000000FFFFFFFFFFFFFFFF"
             "BCE6FAADA7179E84F3B9CAC2FC632551", 16)
```

```
def document_text(uid=UID):
    """Line 1: Transfer Rs.<A> to Account <last 4 digits of UID>, A = 1000 + (UID mod 9000)
       Line 2: Issued by UID: <UID>"""
    amount = 1000 + (int(uid) % 9000)
    return f"Transfer Rs.{amount} to Account {uid[-4:]}\nIssued by UID: {uid}\n"


def ecdsa_private_scalar(uid=UID):
    """d = ( int( SHA-256( UID || "CNS-DSIG-ECDSA-KEY" ) ) mod (n - 1) ) + 1"""
    digest = hashlib.sha256((uid + "CNS-DSIG-ECDSA-KEY").encode()).digest()
    return (int.from_bytes(digest, "big") % (P256_N - 1)) + 1


def bench_message(uid=UID):
    """1024 bytes produced by random.Random(int(UID)).randbytes(1024)"""
    return random.Random(int(uid)).randbytes(1024)


def tamper_index(length, uid=UID):
    """index = UID mod (length of document or signature in bytes)"""
    return int(uid) % length


# ---------------------------------------------------------------- file paths
KEYS_DIR, OUTBOX_DIR, INBOX_DIR = "keys", "outbox", "inbox"


def private_key_path(alg):
    return os.path.join(KEYS_DIR, f"{alg}_private.key")


def public_key_path(alg, folder=OUTBOX_DIR):
    return os.path.join(folder, f"{alg}_public.key")


def signature_path(alg, folder=OUTBOX_DIR):
    return os.path.join(folder, f"{alg}.sig")


def document_path(folder=OUTBOX_DIR):
    return os.path.join(folder, "document.txt")


def read_bytes(path):
    with open(path, "rb") as f:
        return f.read()


def write_bytes(path, data):
    os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
    with open(path, "wb") as f:
        f.write(data)


# ---------------------------------------------------------------- timing helper
def timed_ms(func, *args, **kwargs):
    """Run func once; return (result, elapsed milliseconds) measured with perf_counter_ns."""
    t0 = time.perf_counter_ns()
    result = func(*args, **kwargs)
    t1 = time.perf_counter_ns()
    return result, (t1 - t0) / 1e6
```

keygen.py

```
"""keygen.py <alg> [--independent] - generate the key pair of the chosen algorithm (Step D3).

--independent (wrong-key test, Part H Test 4 only): the ECDSA key comes from the secure random
generator instead of the UID. Without it, the receiver would re-derive the sender's exact ECDSA
key from the same STUDENT_UID, and the "wrong" key would really be the right one.

Private key -> keys/<alg>_private.key   (never leaves the container)
Public key  -> outbox/<alg>_public.key
```

```
"""
import sys

import oqs
from cryptography.hazmat.primitives import serialization
from cryptography.hazmat.primitives.asymmetric import ec, rsa

from common import (ALGS, check_alg, ecdsa_private_scalar, private_key_path,
                    public_key_path, timed_ms, write_bytes)


def _der_pair(private_key):
    sk = private_key.private_bytes(encoding=serialization.Encoding.DER,
                                   format=serialization.PrivateFormat.PKCS8,
                                   encryption_algorithm=serialization.NoEncryption())
    pk = private_key.public_key().public_bytes(
        encoding=serialization.Encoding.DER,
        format=serialization.PublicFormat.SubjectPublicKeyInfo)
    return sk, pk


def generate(alg, independent=False):
    """Return (private_key_bytes, public_key_bytes) for alg."""
    if alg == "ecdsa" and independent:
        return _der_pair(ec.generate_private_key(ec.SECP256R1()))
    if alg == "ecdsa":
        # private key built from the UID-derived scalar d on curve P-256
        return _der_pair(ec.derive_private_key(ecdsa_private_scalar(), ec.SECP256R1()))
    if alg == "rsapss":
        p = ALGS["rsapss"]
        return _der_pair(rsa.generate_private_key(public_exponent=p["e"], key_size=p["key_size"]))
    # ML-DSA / SLH-DSA: raw public key and raw secret key from liboqs
    with oqs.Signature(ALGS[alg]["oqs_name"]) as signer:
        pk = signer.generate_keypair()
        sk = signer.export_secret_key()
    return sk, pk


def main():
    if len(sys.argv) not in (2, 3) or (len(sys.argv) == 3 and sys.argv[2] != "--independent"):
        sys.exit("usage: python3 keygen.py <ecdsa|rsapss|mldsa|slhdsa> [--independent]")
    alg = check_alg(sys.argv[1])
    independent = len(sys.argv) == 3
    (sk, pk), ms = timed_ms(generate, alg, independent)
    write_bytes(private_key_path(alg), sk)
    write_bytes(public_key_path(alg), pk)
    name = ALGS[alg].get("oqs_name", ALGS[alg]["label"])
    print(f"[keygen] algorithm      : {alg} ({name})"
          + ("  [independent key from the secure RNG]" if independent and alg == "ecdsa" else ""))
    print(f"[keygen] private key    : {private_key_path(alg):28s} {len(sk):6d} bytes")
    print(f"[keygen] public key     : {public_key_path(alg):28s} {len(pk):6d} bytes")
    print(f"[keygen] generation time: {ms:.3f} ms")


if __name__ == "__main__":
    main()
```

sign.py

```
"""sign.py <alg> - sign the UID document (Part E).

(1) create outbox/document.txt from the UID if it does not exist, (2) read it as bytes,
(3) load the private key, (4) sign, (5) write outbox/<alg>.sig, (6) print the signature size.
"""
import os
import sys

import oqs
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import ec, padding
from cryptography.hazmat.primitives.serialization import load_der_private_key

from common import (ALGS, check_alg, document_path, document_text, private_key_path,
```

```
                    read_bytes, signature_path, timed_ms, write_bytes)


def load_private(alg, sk_bytes):
    """Classical: cryptography key object. PQ: liboqs signer holding the secret key."""
    if ALGS[alg]["family"] == "classical":
        return load_der_private_key(sk_bytes, password=None)
    return oqs.Signature(ALGS[alg]["oqs_name"], sk_bytes)


def sign_data(alg, key, data):
    if alg == "ecdsa":
        # SHA-256 with ECDSA, DER-encoded signature
        return key.sign(data, ec.ECDSA(hashes.SHA256()))
    if alg == "rsapss":
        # PSS, MGF1(SHA-256), salt length = SHA-256 digest length (32), hash SHA-256
        return key.sign(data,
                        padding.PSS(mgf=padding.MGF1(hashes.SHA256()),
                                    salt_length=ALGS["rsapss"]["salt_length"]),
                        hashes.SHA256())
    # ML-DSA / SLH-DSA: sign the whole document directly (no manual hashing)
    return key.sign(data)


def main():
    if len(sys.argv) != 2:
        sys.exit("usage: python3 sign.py <ecdsa|rsapss|mldsa|slhdsa>")
    alg = check_alg(sys.argv[1])

    doc = document_path()
    if not os.path.exists(doc):
        write_bytes(doc, document_text().encode())
        print(f"[sign] created {doc}")
    data = read_bytes(doc)

    key = load_private(alg, read_bytes(private_key_path(alg)))
    sig, ms = timed_ms(sign_data, alg, key, data)
    write_bytes(signature_path(alg), sig)

    print(f"[sign] algorithm     : {alg}")
    print(f"[sign] document      : {doc} ({len(data)} bytes)")
    print(f"[sign] signature     : {signature_path(alg)} ({len(sig)} bytes)")
    print(f"[sign] signing time  : {ms:.3f} ms")


if __name__ == "__main__":
    main()
```

verify.py

```
"""verify.py <alg> [document] [public_key] [signature] - verify a signature (Part G).

Defaults: inbox/document.txt, inbox/<alg>_public.key, inbox/<alg>.sig
Uses exactly the same parameters as sign.py; prints PASS or FAIL and the verification time.
"""
import sys

import oqs
from cryptography.exceptions import InvalidSignature
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.asymmetric import ec, padding
from cryptography.hazmat.primitives.serialization import load_der_public_key

from common import (ALGS, INBOX_DIR, check_alg, document_path, public_key_path,
                    read_bytes, signature_path, timed_ms)


def load_public(alg, pk_bytes):
    """Classical: cryptography public-key object. PQ: (liboqs verifier, raw public key)."""
    if ALGS[alg]["family"] == "classical":
        return load_der_public_key(pk_bytes)
    return oqs.Signature(ALGS[alg]["oqs_name"]), pk_bytes
```

```
def verify_data(alg, key, data, sig):
    """Return True if the signature is valid, False otherwise."""
    if ALGS[alg]["family"] == "classical":
        try:
            if alg == "ecdsa":
                key.verify(sig, data, ec.ECDSA(hashes.SHA256()))
            else:
                key.verify(sig, data,
                           padding.PSS(mgf=padding.MGF1(hashes.SHA256()),
                                       salt_length=ALGS["rsapss"]["salt_length"]),
                           hashes.SHA256())
            return True
        except InvalidSignature:          # cryptography raises on failure
            return False
    verifier, pk = key
    return verifier.verify(data, sig, pk)  # liboqs returns a boolean


def main():
    if not 2 <= len(sys.argv) <= 5:
        sys.exit("usage: python3 verify.py <alg> [document] [public_key] [signature]")
    alg = check_alg(sys.argv[1])
    doc = sys.argv[2] if len(sys.argv) > 2 else document_path(INBOX_DIR)
    pub = sys.argv[3] if len(sys.argv) > 3 else public_key_path(alg, INBOX_DIR)
    sig_file = sys.argv[4] if len(sys.argv) > 4 else signature_path(alg, INBOX_DIR)

    data, sig = read_bytes(doc), read_bytes(sig_file)
    try:
        key = load_public(alg, read_bytes(pub))
        ok, ms = timed_ms(verify_data, alg, key, data, sig)
    except ValueError as exc:             # e.g. malformed key / wrong key length
        ok, ms = False, 0.0
        print(f"[verify] error: {exc}")

    print(f"[verify] algorithm : {alg}")
    print(f"[verify] document  : {doc}")
    print(f"[verify] public key: {pub}")
    print(f"[verify] signature : {sig_file}")
    print(f"[verify] time      : {ms:.3f} ms")
    print(f"[verify] RESULT    : {'PASS (VALID)' if ok else 'FAIL (INVALID)'}")
    sys.exit(0 if ok else 1)


if __name__ == "__main__":
    main()
```

tamper.py

```
"""tamper.py <doc|sig> <file> - Mallory modifies one character/bit (Part H).

index = UID mod (file length); the byte at that index is XORed with 0x01 (exactly one bit,
hence exactly one character, changes). The result is saved as <file>.tampered; the original
file is never overwritten.
"""
import sys

from common import read_bytes, tamper_index, write_bytes


def main():
    if len(sys.argv) != 3 or sys.argv[1] not in ("doc", "sig"):
        sys.exit("usage: python3 tamper.py <doc|sig> <file>")
    kind, path = sys.argv[1], sys.argv[2]
    data = bytearray(read_bytes(path))
    idx = tamper_index(len(data))
    old = data[idx]
    data[idx] ^= 0x01
    out = path + ".tampered"
    write_bytes(out, bytes(data))

    print(f"[tamper] file      : {path} ({len(data)} bytes)")
    print(f"[tamper] index     : UID mod {len(data)} = {idx}")
```

```
    print(f"[tamper] byte      : 0x{old:02x} -> 0x{data[idx]:02x} (bit 0 flipped)")
    if kind == "doc":
        print(f"[tamper] character : {chr(old)!r} -> {chr(data[idx])!r}")
        print("[tamper] original text:")
        print(read_bytes(path).decode(errors="replace"), end="")
        print("[tamper] modified text:")
        print(bytes(data).decode(errors="replace"), end="")
    print(f"[tamper] written   : {out}")


if __name__ == "__main__":
    main()
```

bench.py

```
"""bench.py [output.csv] - benchmark key generation, signing and verification (Part I).

Only the cryptographic call is inside the timed region (no Docker start-up, file I/O or
network). Per algorithm and operation: 5 un-recorded warm-up runs, then key generation x20,
signing x100, verification x100, each call timed individually with time.perf_counter_ns().
The message is the 1024-byte UID-seeded message from Section 5.
"""
import csv
import platform
import socket
import statistics
import sys
import time

from common import ALG_ORDER, ALGS, UID, bench_message
from keygen import generate
from sign import load_private, sign_data
from verify import load_public, verify_data

WARMUP = 5
RUNS = {"keygen": 20, "sign": 100, "verify": 100}


def measure(func, runs):
    for _ in range(WARMUP):                      # warm-up, not recorded
        func()
    times, result = [], None
    for _ in range(runs):
        t0 = time.perf_counter_ns()
        result = func()
        t1 = time.perf_counter_ns()
        times.append((t1 - t0) / 1e6)            # ns -> ms
    return times, result


def stats(times):
    return {"mean_ms": statistics.mean(times), "std_ms": statistics.stdev(times),
            "min_ms": min(times), "max_ms": max(times)}


def main():
    out = sys.argv[1] if len(sys.argv) > 1 else "results.csv"
    msg = bench_message()
    assert len(msg) == 1024
    print(f"[bench] UID={UID} host={socket.gethostname()} python={platform.python_version()}")
    print(f"[bench] message: 1024 bytes, sha256-prefix={__import__('hashlib').sha256(msg).hexdigest()[:16]}")

    rows = []
    for alg in ALG_ORDER:
        # key generation (the key pair produced last is used for signing/verification)
        kg_times, (sk, pk) = measure(lambda: generate(alg), RUNS["keygen"])
        priv, pub = load_private(alg, sk), load_public(alg, pk)

        sg_times, sig = measure(lambda: sign_data(alg, priv, msg), RUNS["sign"])
        vf_times, ok = measure(lambda: verify_data(alg, pub, msg, sig), RUNS["verify"])

        # the signature from the last signing loop must verify, otherwise we timed failures
        if not (ok and verify_data(alg, pub, msg, sig)):
```

```
            sys.exit(f"ERROR: {alg} signature from the benchmark did not verify")

        for op, times in (("keygen", kg_times), ("sign", sg_times), ("verify", vf_times)):
            s = stats(times)
            rows.append({"uid": UID, "algorithm": alg, "operation": op, "runs": len(times),
                         **{k: f"{v:.4f}" for k, v in s.items()},
                         "public_key_bytes": len(pk), "private_key_bytes": len(sk),
                         "signature_bytes": len(sig)})
            print(f"[bench] {alg:7s} {op:7s} runs={len(times):3d} mean={s['mean_ms']:10.4f} ms "
                  f"std={s['std_ms']:9.4f} min={s['min_ms']:10.4f} max={s['max_ms']:10.4f}")
        print(f"[bench] {alg:7s} sizes: pk={len(pk)} B  sk={len(sk)} B  sig={len(sig)} B  "
              f"verify-check=PASS")

    fields = ["uid", "algorithm", "operation", "runs", "mean_ms", "std_ms", "min_ms", "max_ms",
              "public_key_bytes", "private_key_bytes", "signature_bytes"]
    with open(out, "w", newline="") as f:
        w = csv.DictWriter(f, fieldnames=fields)
        w.writeheader()
        w.writerows(rows)
    print(f"[bench] written {out}")


if __name__ == "__main__":
    main()
```

plot.py

```
"""plot.py [results.csv] [document.txt] - draw the seven graphs of Section 15 into graphs/.

Every graph has a title, axis labels with units, a legend where required and the execution
environment in the caption (read from environment.txt if present). A logarithmic y-axis is
used wherever the values differ by orders of magnitude.
"""
import os
import platform
import sys

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

from common import ALG_ORDER, ALGS, UID

CSV = sys.argv[1] if len(sys.argv) > 1 else "results.csv"
DOC = sys.argv[2] if len(sys.argv) > 2 else os.path.join("outbox", "document.txt")
OUT = "graphs"
COLORS = {"ecdsa": "#2a6fb0", "rsapss": "#7a4fa3", "mldsa": "#2e8b57", "slhdsa": "#d0782a"}


def environment_caption():
    if os.path.exists("environment.txt"):
        with open("environment.txt") as f:
            env = " | ".join(line.strip() for line in f if line.strip())
    else:
        env = f"{platform.processor() or platform.machine()} | {platform.platform()}"
    return f"UID {UID} | Environment: {env}"


CAPTION = environment_caption()

df = pd.read_csv(CSV, dtype={"uid": str})
labels = [ALGS[a]["label"] for a in ALG_ORDER]
colors = [COLORS[a] for a in ALG_ORDER]
x = np.arange(len(ALG_ORDER))


def row(alg, op):
    return df[(df.algorithm == alg) & (df.operation == op)].iloc[0]


def sizes(col):
```

```
    return [int(row(a, "sign")[col]) for a in ALG_ORDER]


pk, sk, sig = sizes("public_key_bytes"), sizes("private_key_bytes"), sizes("signature_bytes")
doc = os.path.getsize(DOC)


def finish(fig, ax, name, title, ylabel, log=False, legend=False):
    ax.set_title(title, fontsize=12, fontweight="bold")
    ax.set_xlabel("Signature algorithm")
    ax.set_ylabel(ylabel)
    ax.set_xticks(x)
    ax.set_xticklabels(labels)
    if log:
        ax.set_yscale("log")
    ax.grid(axis="y", which="both", alpha=0.3)
    ax.set_axisbelow(True)
    if legend:
        ax.legend()
    fig.text(0.5, 0.01, CAPTION, ha="center", va="bottom", fontsize=6.5, wrap=True, color="#444")
    fig.tight_layout(rect=(0, 0.06, 1, 1))
    path = os.path.join(OUT, name)
    fig.savefig(path, dpi=150)
    plt.close(fig)
    print(f"[plot] wrote {path}")


def annotate(ax, bars, fmt="{:,.0f}", tops=None):
    for i, b in enumerate(bars):
        h = b.get_height()
        y = tops[i] if tops is not None else h
        ax.annotate(fmt.format(h), (b.get_x() + b.get_width() / 2, y), ha="center",
                    va="bottom", fontsize=8, xytext=(0, 2), textcoords="offset points")


def bar_with_err(op, name, title):
    mean = [row(a, op).mean_ms for a in ALG_ORDER]
    std = [row(a, op).std_ms for a in ALG_ORDER]
    fig, ax = plt.subplots(figsize=(8, 5))
    bars = ax.bar(x, mean, yerr=std, capsize=6, color=colors, edgecolor="black",
                  error_kw={"elinewidth": 1.2})
    annotate(ax, bars, "{:.3f}", tops=[m + sd for m, sd in zip(mean, std)])
    n = int(row(ALG_ORDER[0], op).runs)
    finish(fig, ax, name, title, f"Mean time per operation (ms, log scale), n={n}", log=True)


def main():
    os.makedirs(OUT, exist_ok=True)

    # Graph 1 - grouped bar: public and private key sizes
    fig, ax = plt.subplots(figsize=(8, 5))
    w = 0.38
    b1 = ax.bar(x - w / 2, pk, w, label="Public key", color="#2a6fb0", edgecolor="black")
    b2 = ax.bar(x + w / 2, sk, w, label="Private key", color="#d0782a", edgecolor="black")
    annotate(ax, b1); annotate(ax, b2)
    finish(fig, ax, "graph1_key_sizes.png", "Graph 1: Public- and private-key size",
           "Size (bytes, log scale)", log=True, legend=True)

    # Graph 2 - bar: signature size
    fig, ax = plt.subplots(figsize=(8, 5))
    annotate(ax, ax.bar(x, sig, color=colors, edgecolor="black"))
    finish(fig, ax, "graph2_signature_size.png", "Graph 2: Signature size",
           "Signature size (bytes, log scale)", log=True)

    # Graphs 3, 4 - bars with error bars (std)
    bar_with_err("sign", "graph3_signing_time.png",
                 "Graph 3: Average signing time (1024-byte message), error bar = std")
    bar_with_err("verify", "graph4_verification_time.png",
                 "Graph 4: Average verification time (1024-byte message), error bar = std")

    # Graph 5 - bar: key-generation time
    fig, ax = plt.subplots(figsize=(8, 5))
    kg = [row(a, "keygen").mean_ms for a in ALG_ORDER]
```

```
    annotate(ax, ax.bar(x, kg, color=colors, edgecolor="black"), "{:.3f}")
    finish(fig, ax, "graph5_keygen_time.png", "Graph 5: Average key-generation time",
           f"Mean time per key pair (ms, log scale), n={int(row('ecdsa', 'keygen').runs)}", log=True)

    # Graph 6 - stacked bar: document + signature + public key
    fig, ax = plt.subplots(figsize=(8, 5))
    d = np.array([doc] * len(ALG_ORDER)); s = np.array(sig); p = np.array(pk)
    ax.bar(x, d, label=f"Document ({doc} B)", color="#999999", edgecolor="black")
    ax.bar(x, s, bottom=d, label="Signature", color="#d0782a", edgecolor="black")
    bars = ax.bar(x, p, bottom=d + s, label="Public key", color="#2a6fb0", edgecolor="black")
    for b, tot in zip(bars, d + s + p):
        ax.annotate(f"{tot:,} B", (b.get_x() + b.get_width() / 2, tot), ha="center",
                    va="bottom", fontsize=9, xytext=(0, 2), textcoords="offset points")
    finish(fig, ax, "graph6_transmitted_bytes.png",
           "Graph 6: Total transmitted bytes = document + signature + public key",
           "Bytes sent over sig-net (linear scale)", legend=True)

    # Graph 7 (extra) - grouped bar relative to ECDSA
    fig, ax = plt.subplots(figsize=(9, 5))
    w = 0.26
    for i, (op, name, c) in enumerate((("keygen", "Key generation", "#2e8b57"),
                                       ("sign", "Signing", "#d0782a"),
                                       ("verify", "Verification", "#2a6fb0"))):
        base = row("ecdsa", op).mean_ms
        rel = [row(a, op).mean_ms / base for a in ALG_ORDER]
        annotate(ax, ax.bar(x + (i - 1) * w, rel, w, label=name, color=c, edgecolor="black"),
                 "{:.2f}")
    ax.axhline(1.0, color="black", linestyle="--", linewidth=1)
    finish(fig, ax, "graph7_relative_to_ecdsa.png",
           "Graph 7: Time relative to ECDSA (ECDSA = 1)",
           "Relative time (x ECDSA, log scale)", log=True, legend=True)


if __name__ == "__main__":
    main()
```

The same programs are used in the receiver container (receiver/ folder).

9. results.csv files and UID

UID: 2024300032 (first column of every result file)

Sender results.csv (median run = results_run3.csv)

uid	algorithm	operation	runs	mean_ms	std_ms	min_ms	max_ms	pk_bytes	sk_bytes	sig_bytes
2024300032	ecdsa	keygen	20	0.0922	0.0324	0.0795	0.1998	91	138	72
2024300032	ecdsa	sign	100	0.0442	0.0087	0.0395	0.0983	91	138	72
2024300032	ecdsa	verify	100	0.1239	0.0092	0.1119	0.1632	91	138	72
2024300032	rsapss	keygen	20	198.1258	102.1346	38.9473	383.2885	422	1795	384
2024300032	rsapss	sign	100	1.3343	0.0675	1.3006	1.7054	422	1795	384
2024300032	rsapss	verify	100	0.0342	0.0017	0.0333	0.0424	422	1795	384
2024300032	mldsa	keygen	20	0.0250	0.0006	0.0244	0.0270	1312	2560	2420
2024300032	mldsa	sign	100	0.0659	0.0337	0.0307	0.1682	1312	2560	2420
2024300032	mldsa	verify	100	0.0238	0.0021	0.0228	0.0332	1312	2560	2420
2024300032	slhdsa	keygen	20	55.5836	14.4815	47.3222	100.5666	32	64	7856
2024300032	slhdsa	sign	100	471.3847	83.6407	368.0724	857.8316	32	64	7856
2024300032	slhdsa	verify	100	0.4062	0.0153	0.3659	0.4926	32	64	7856

Receiver results.csv

uid	algorithm	operation	runs	mean_ms	std_ms	min_ms	max_ms	pk_bytes	sk_bytes	sig_bytes
2024300032	ecdsa	keygen	20	0.0324	0.0054	0.0302	0.0544	91	138	70
2024300032	ecdsa	sign	100	0.0194	0.0015	0.0188	0.0326	91	138	70
2024300032	ecdsa	verify	100	0.0597	0.0124	0.0532	0.1340	91	138	70
2024300032	rsapss	keygen	20	115.7305	76.3442	49.8620	323.9396	422	1794	384
2024300032	rsapss	sign	100	1.3643	0.1119	1.3034	2.0226	422	1794	384
2024300032	rsapss	verify	100	0.0347	0.0026	0.0332	0.0489	422	1794	384
2024300032	mldsa	keygen	20	0.0279	0.0053	0.0248	0.0490	1312	2560	2420
2024300032	mldsa	sign	100	0.0632	0.0363	0.0308	0.2783	1312	2560	2420
2024300032	mldsa	verify	100	0.0253	0.0042	0.0230	0.0537	1312	2560	2420
2024300032	slhdsa	keygen	20	64.1955	23.6940	47.6752	132.3601	32	64	7856
2024300032	slhdsa	sign	100	488.9761	68.7288	376.9210	809.7748	32	64	7856
2024300032	slhdsa	verify	100	0.3776	0.0156	0.3334	0.4416	32	64	7856

CSV header: uid,algorithm,operation,runs,mean_ms,std_ms,min_ms,max_ms,public_key_bytes,private_key_bytes,signature_bytes.
Files: results_sender.csv, results_receiver.csv (and sender/results_run1.csv, results_run2.csv, results_run3.csv).